

# Module 5

Basic electronic circuits and instrumentation:

Rectifiers and power supplies:

Block diagram description of a dc power supply,

Working of a full wave bridge rectifier,

capacitor filter (no analysis),

working of simple zener voltage regulator.

Amplifiers: Block diagram of Public Address system,

Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response,

Concept of voltage divider biasing.

Electronic Instrumentation:

Block diagram of an electronic instrumentation system,

Working of digital multimeter.

# DC Power supply

Today almost every electronic device needs a DC supply for its smooth operation and they need to be operated within certain power supply limits.

This required DC voltage or DC supply is derived from single phase ac mains.

# DC Power supply

A regulated power supply can convert unregulated an AC (alternating current or voltage) to a constant DC (direct current or voltage).

A regulated power supply is used to ensure that the output remains constant even if the input changes.

A regulated DC power supply is also called as a linear power supply, it is an embedded circuit and consists of various blocks.

# DC Power supply

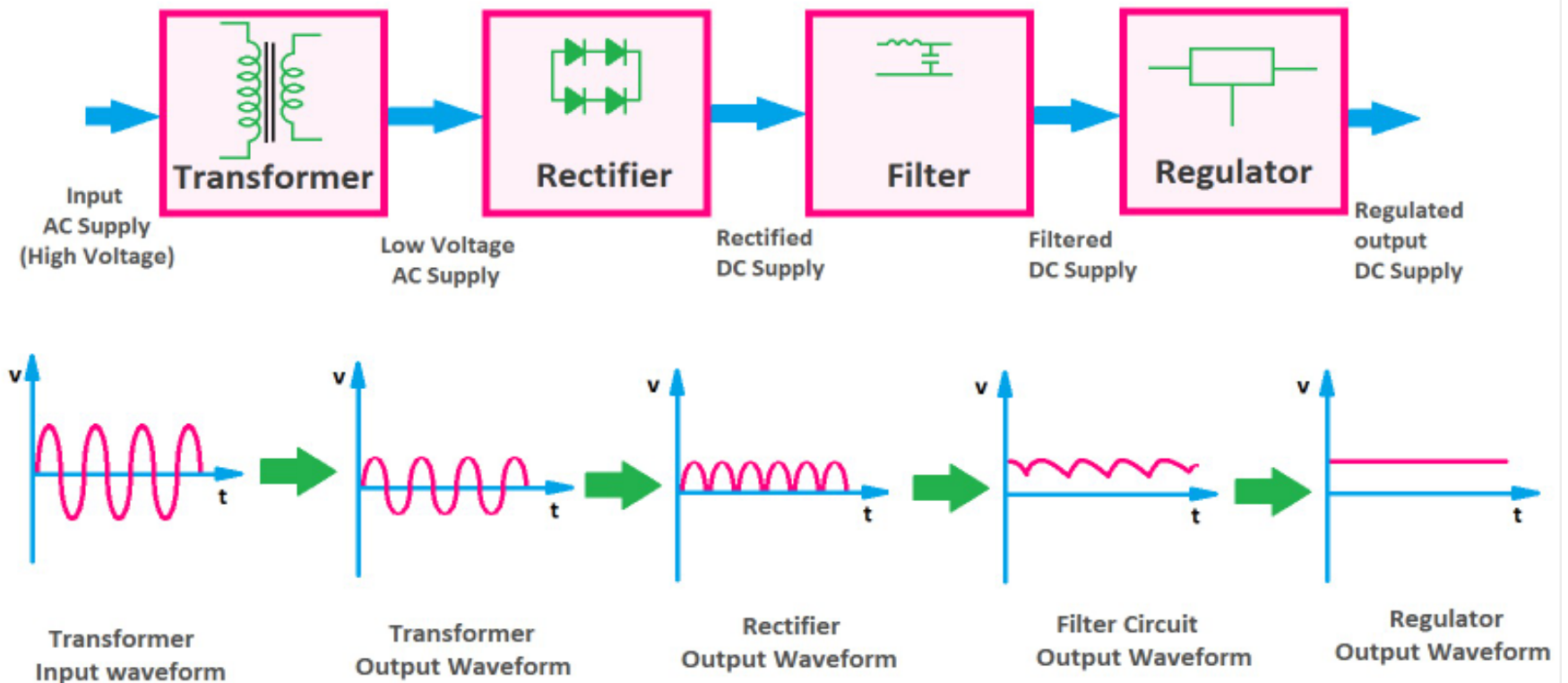
The regulated power supply will accept an AC input and give a constant DC output.

Figure below shows the block diagram of a typical regulated DC power supply

# DC Power supply

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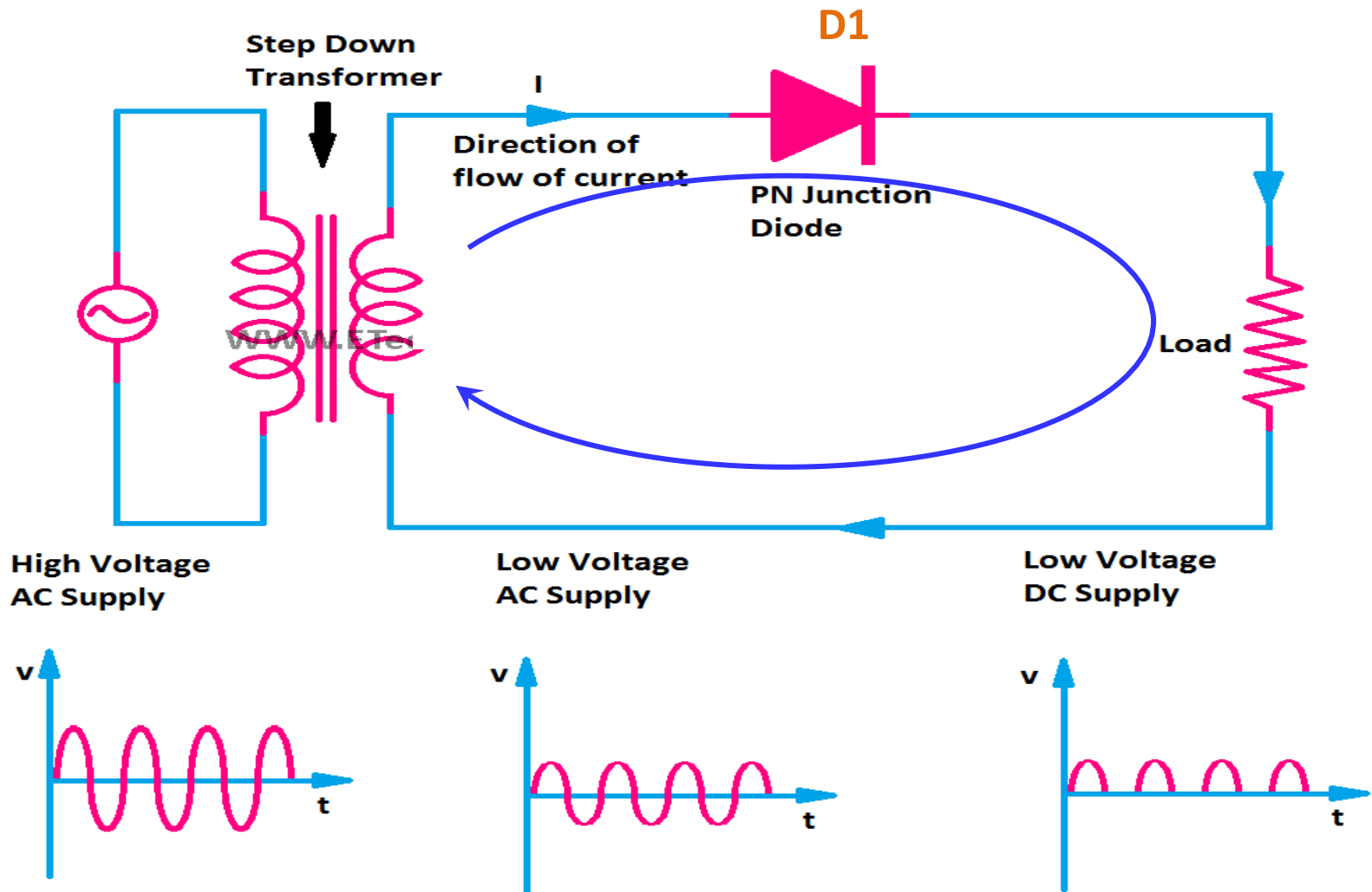
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***Regulated Power Supply Block Diagram***

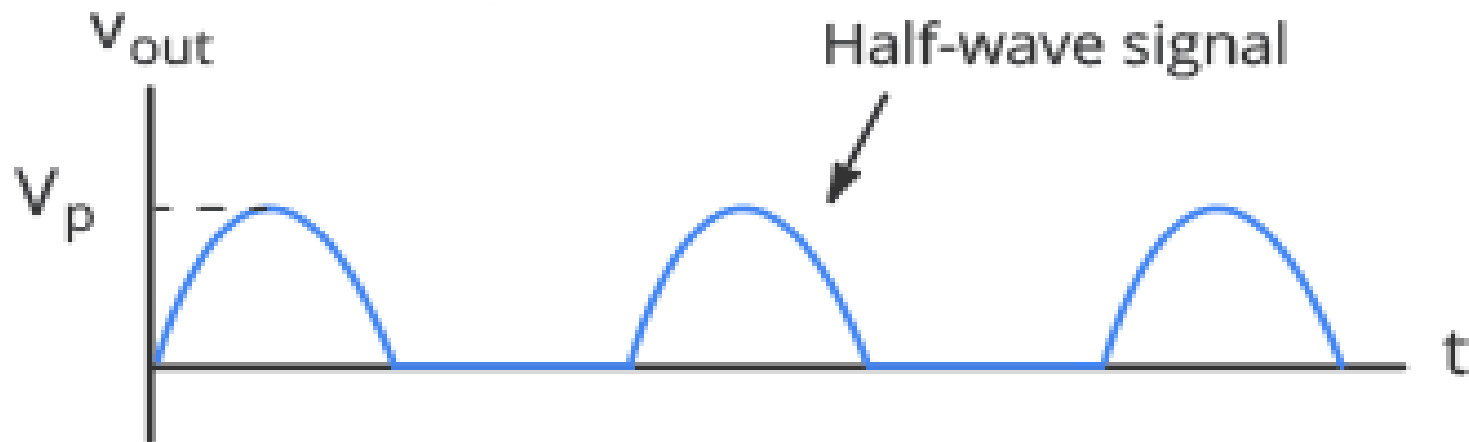
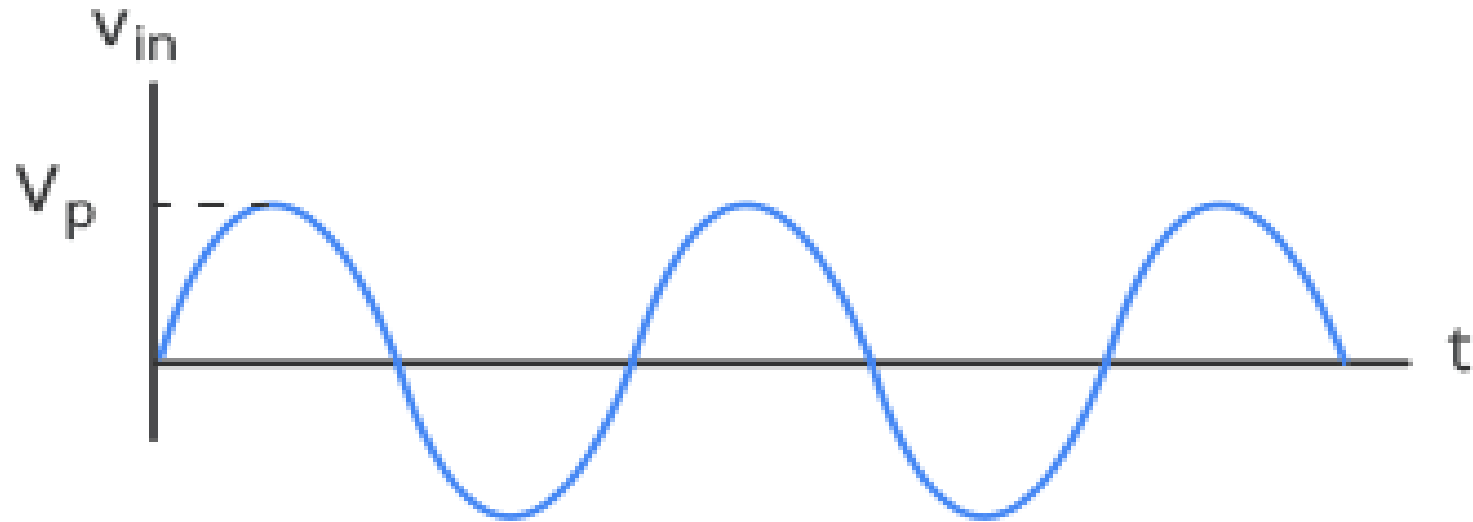
# DC Power supply

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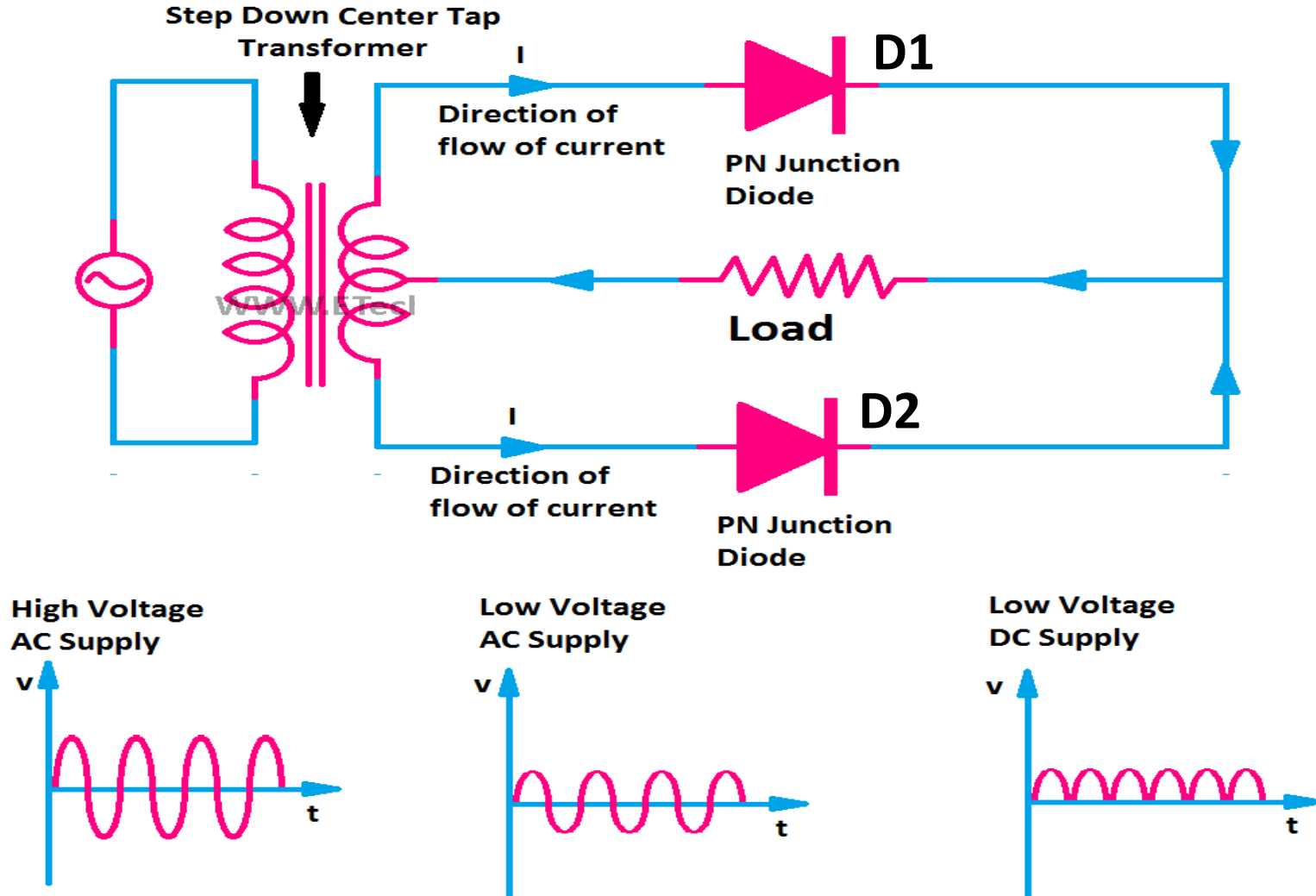
**Half Wave Rectifier Circuit Diagram and Waveform**

# DC Power supply



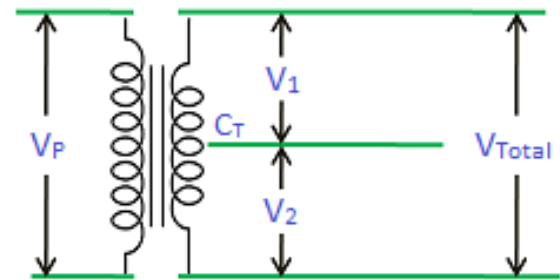
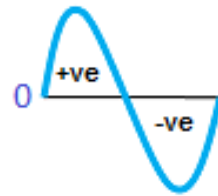
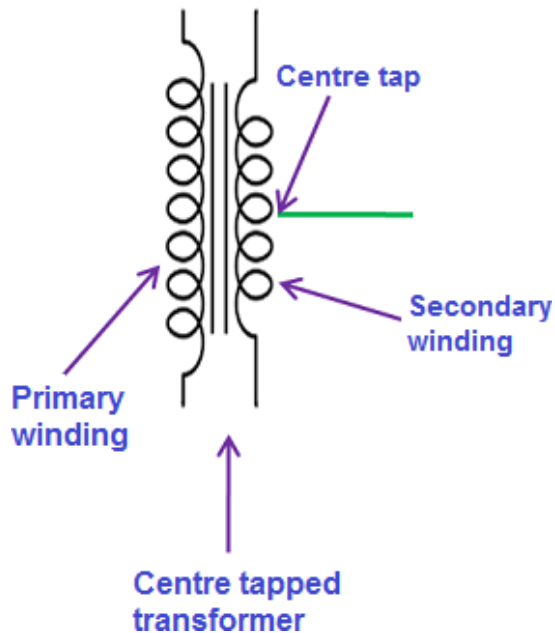


# Centre Tapped FW Rectifier



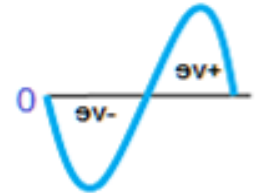
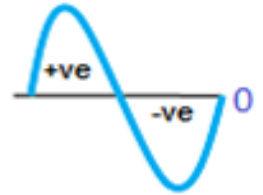
Center Tap Full Wave Rectifier Circuit Diagram and Waveform

# Centre Tapped FW Rectifier

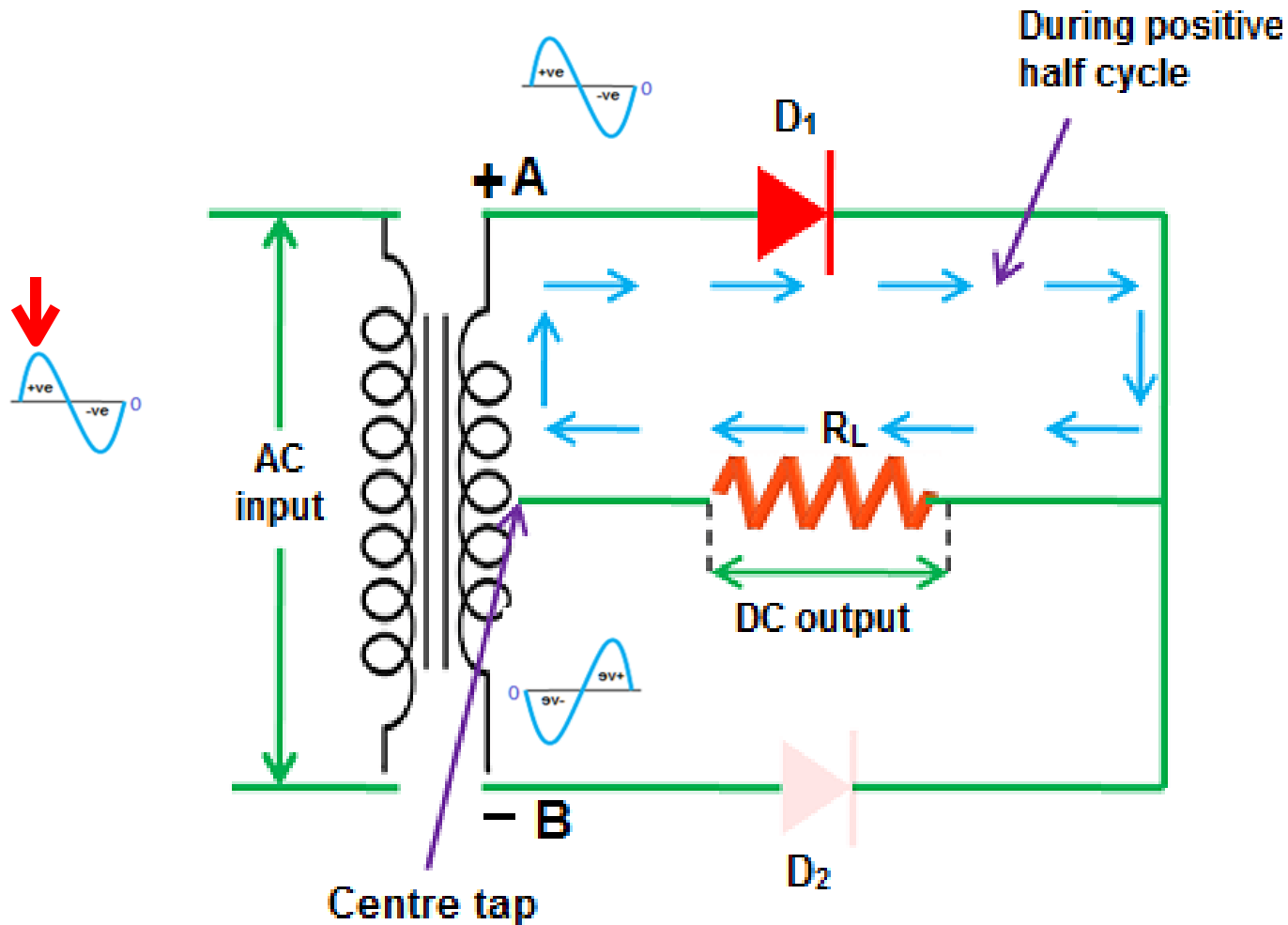


$$V_{Total} = V_1 + V_2$$

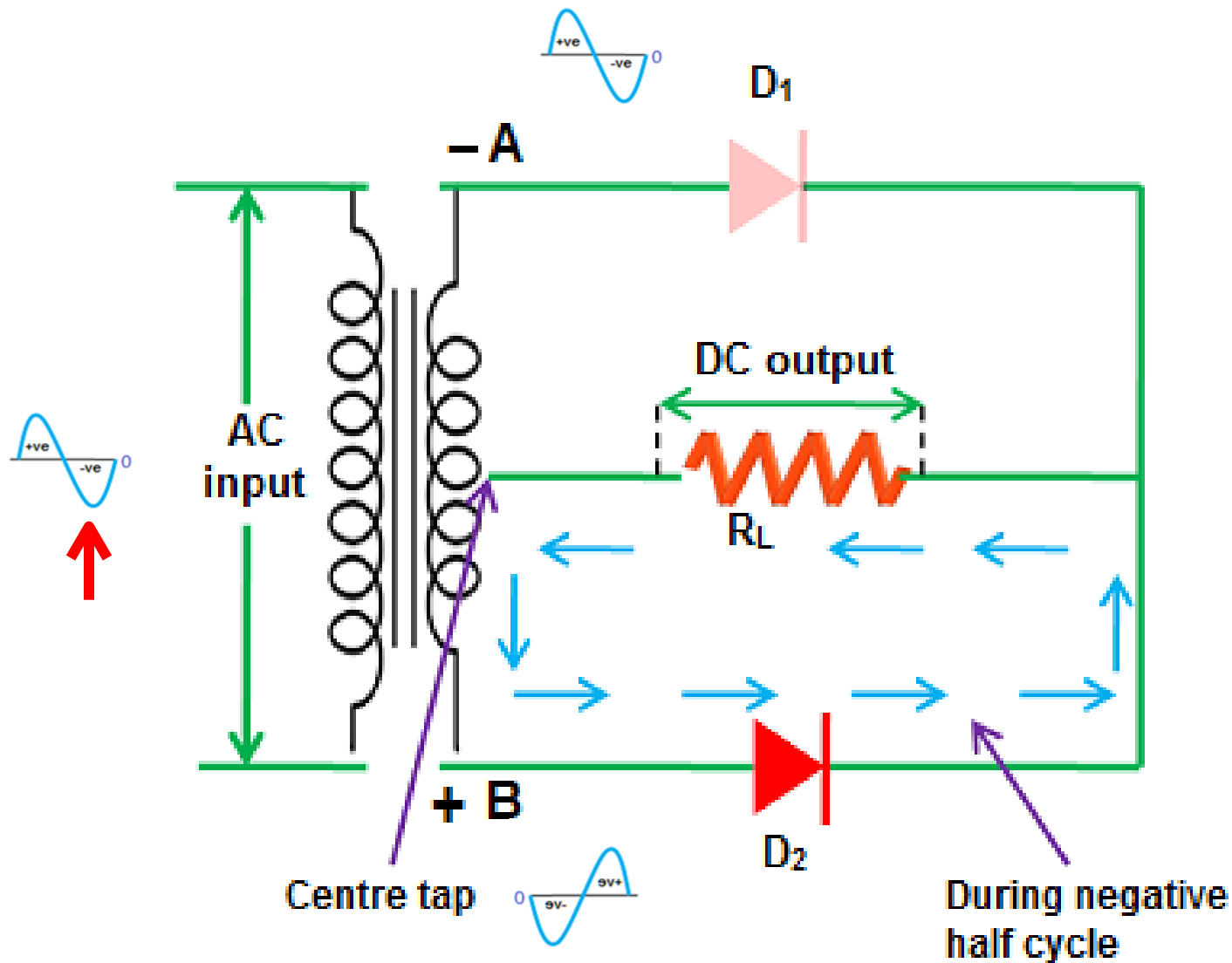
$C_T$  = Centre tap



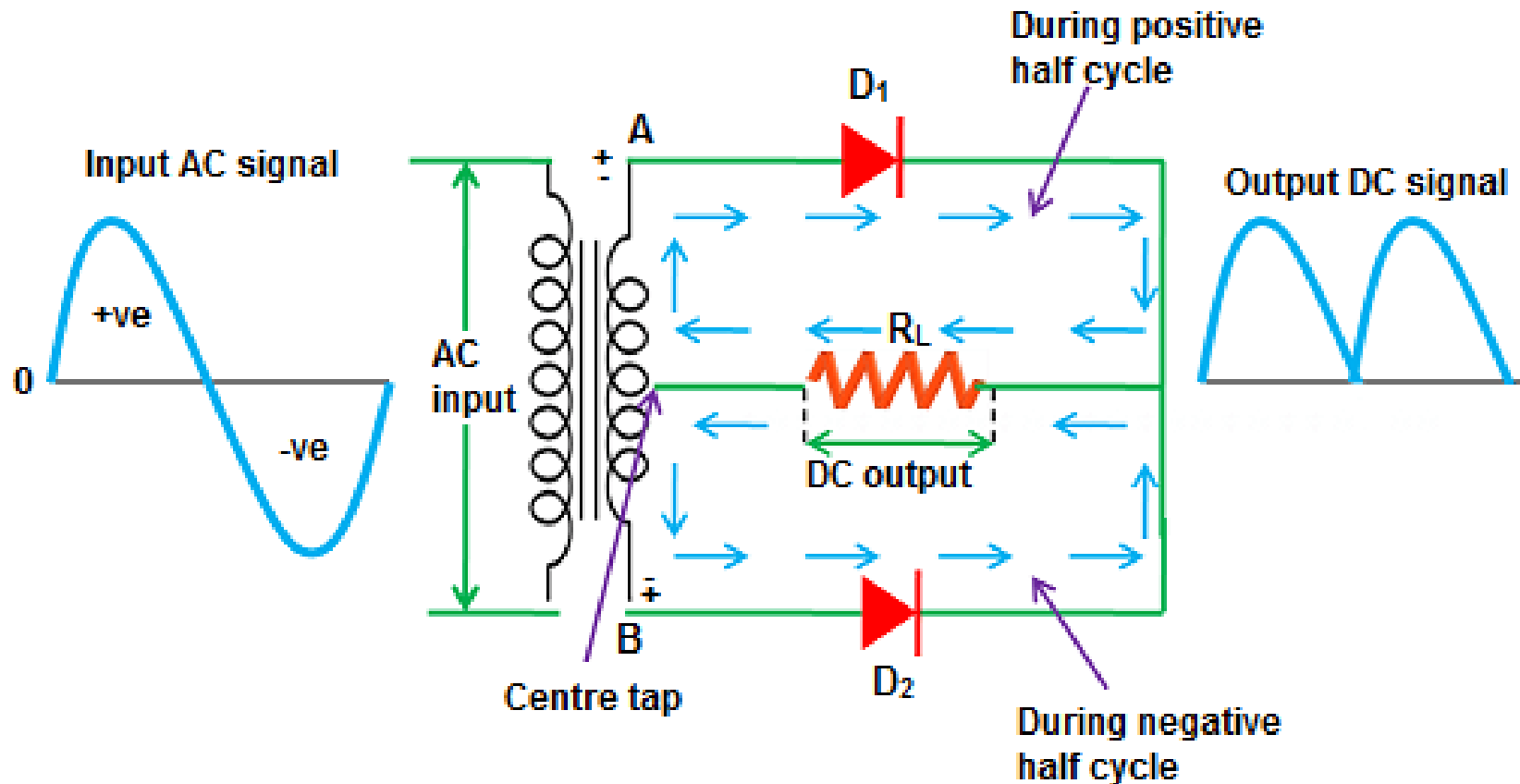
# Centre Tapped FW Rectifier



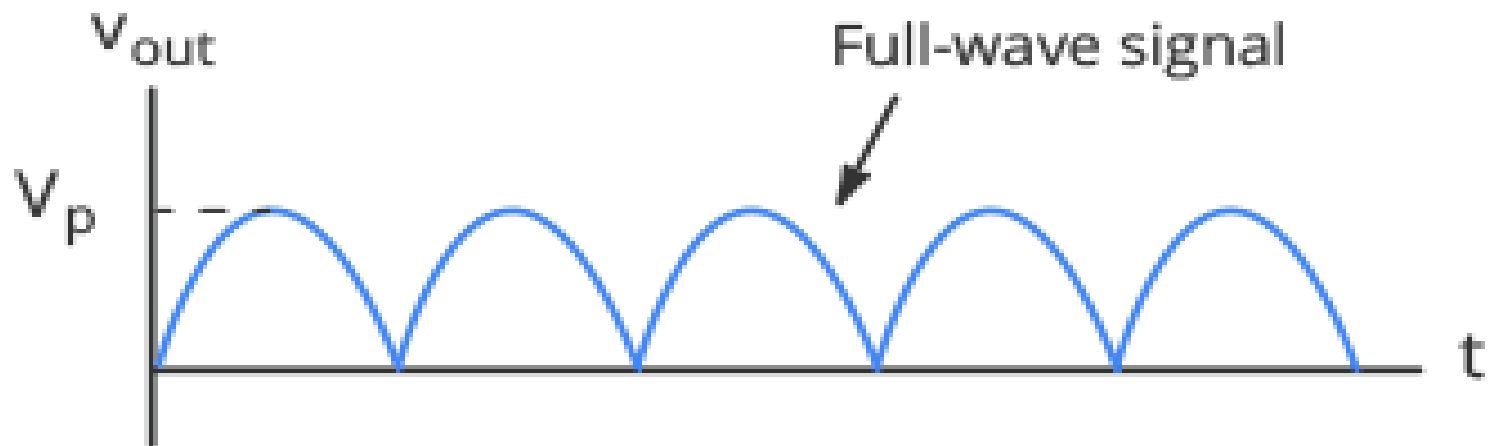
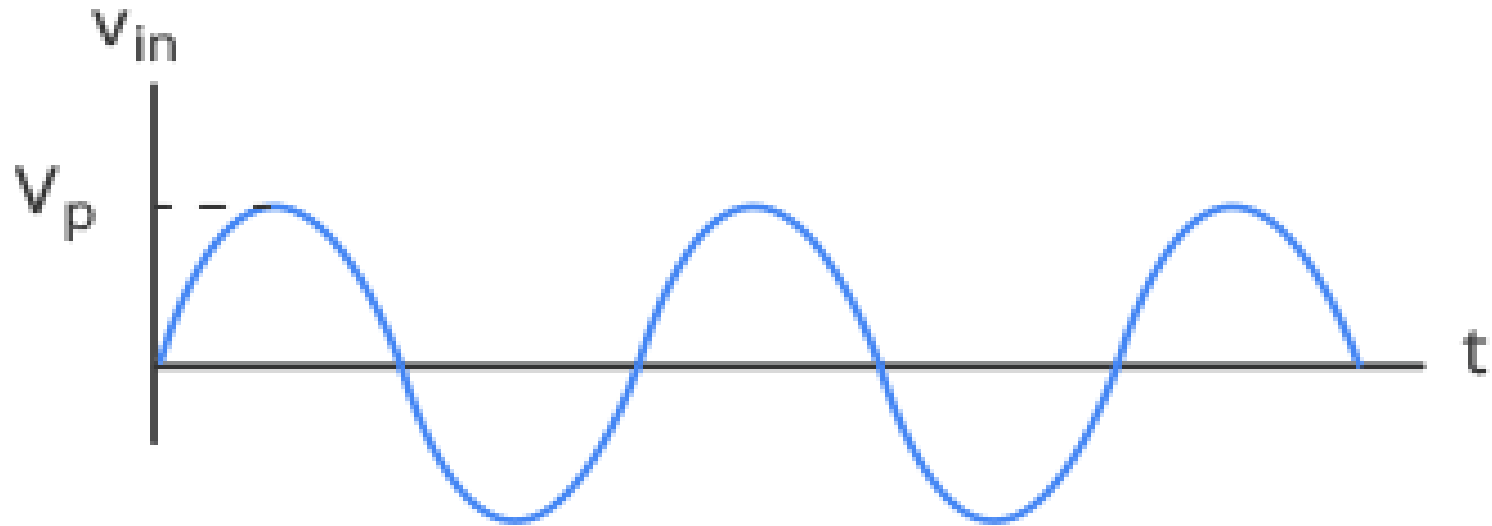
# Centre Tapped FW Rectifier



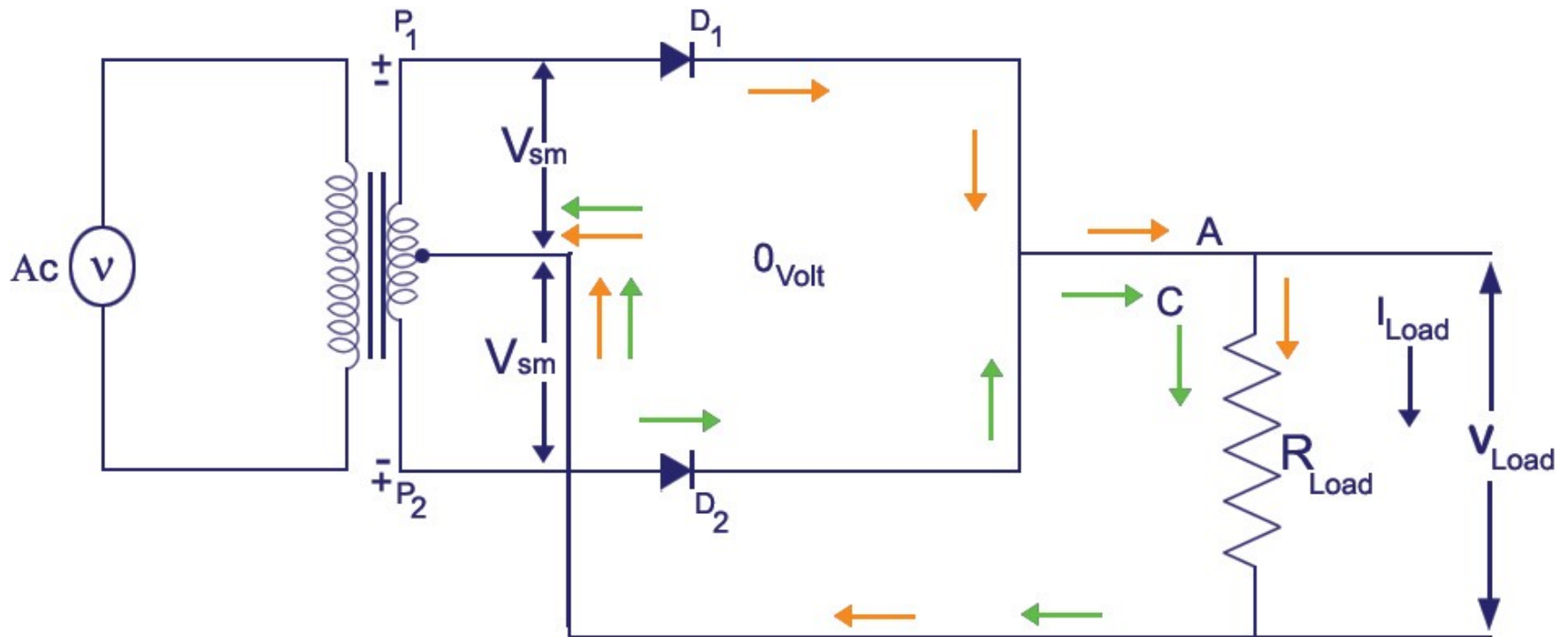
# Centre Tapped FW Rectifier



# CT FW Rectifier Waveforms



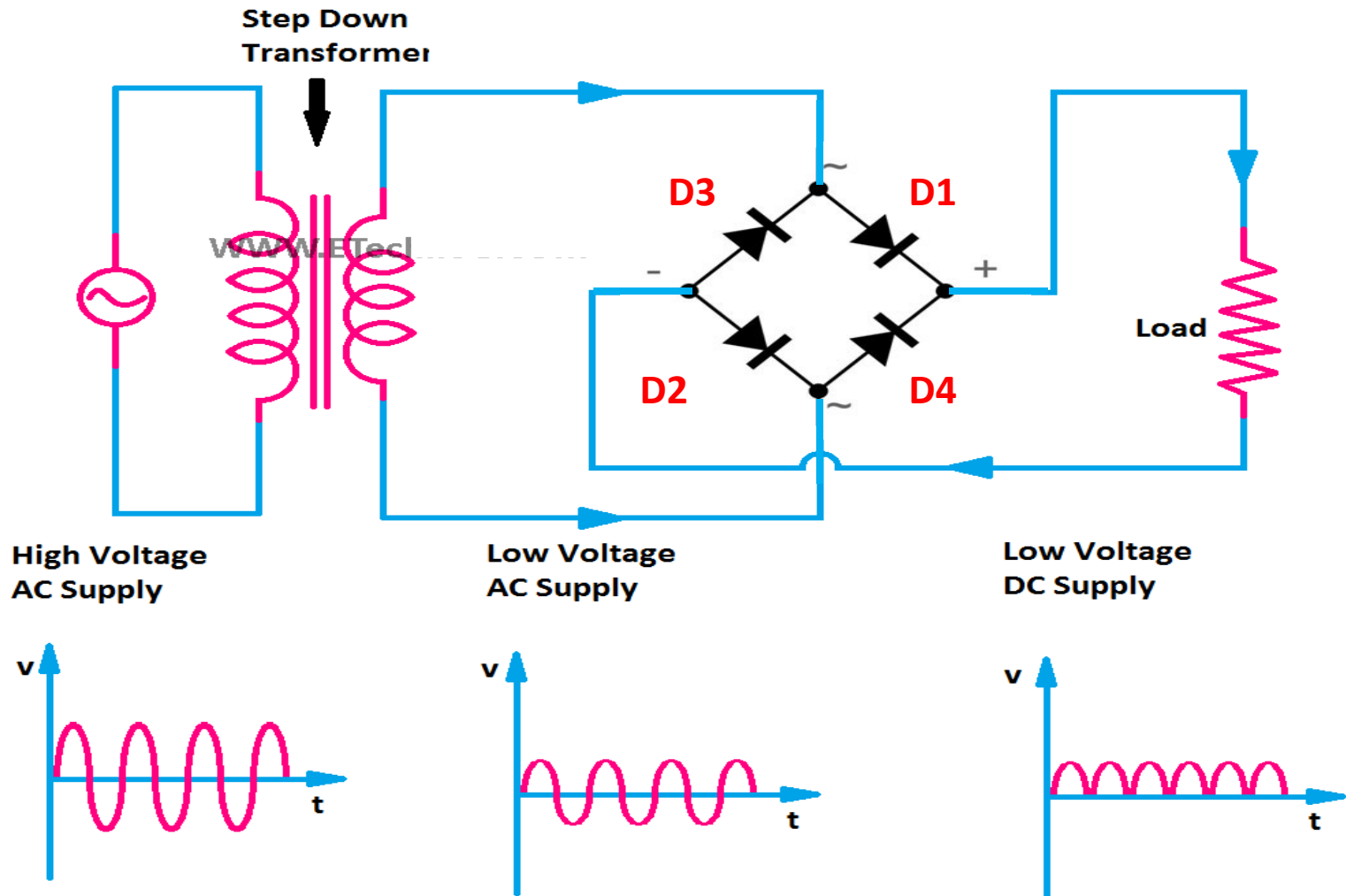
# Centre Tapped FW Rectifier



**CENTRE - TAP FULL- WAVE RECTIFIER CIRCUIT**

# Bridge Rectifier

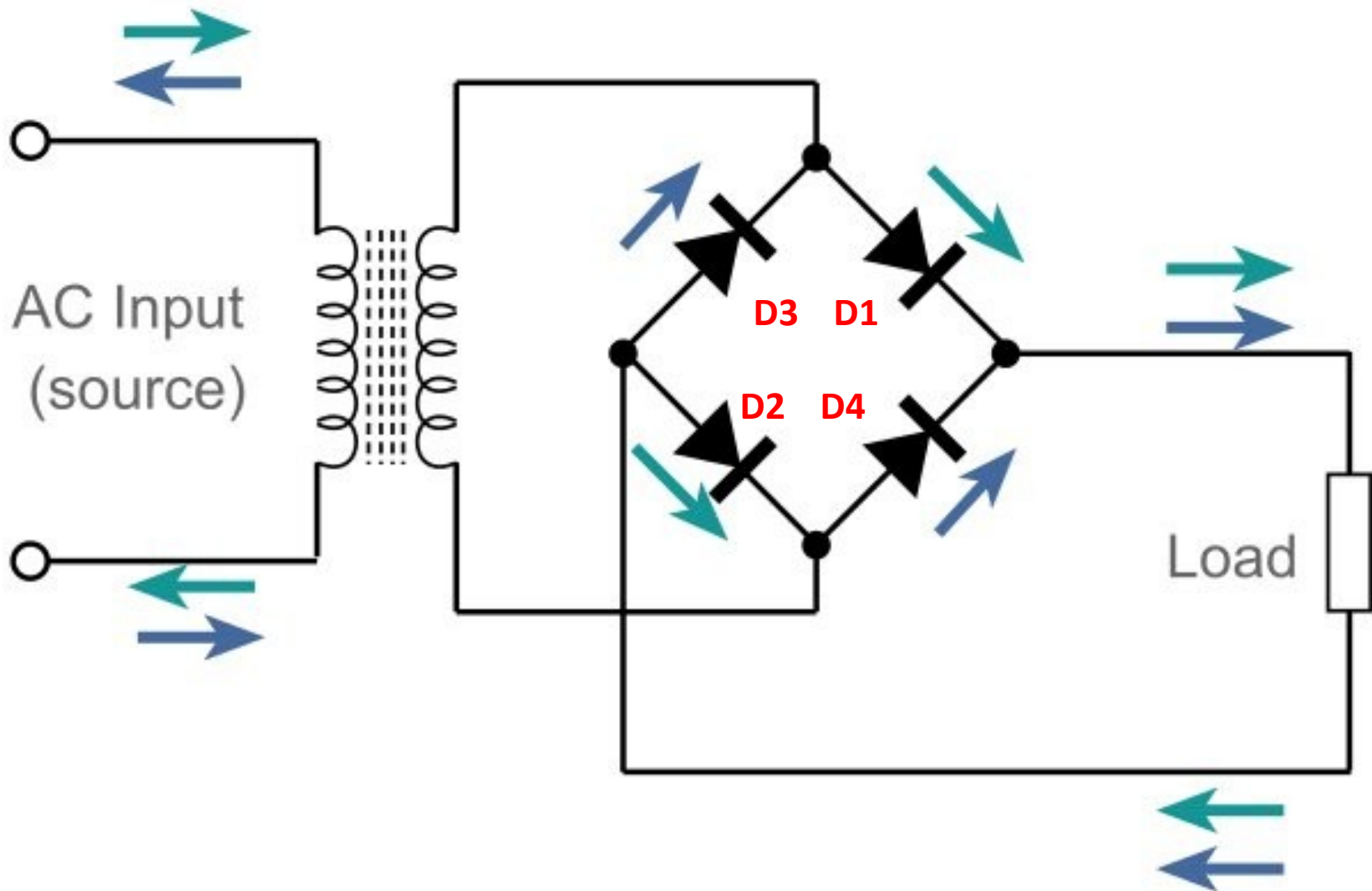
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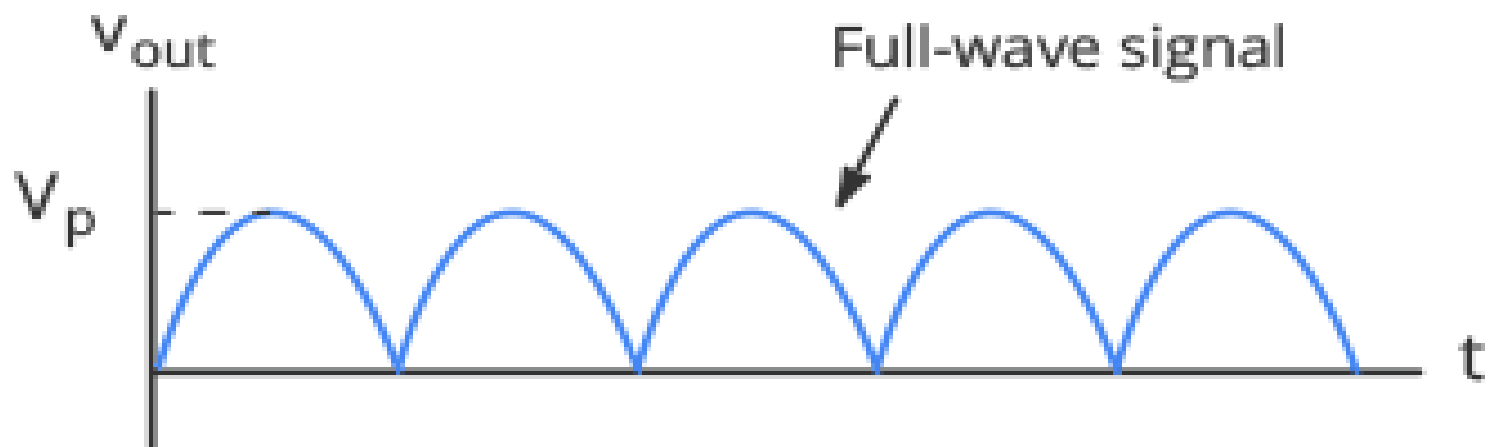
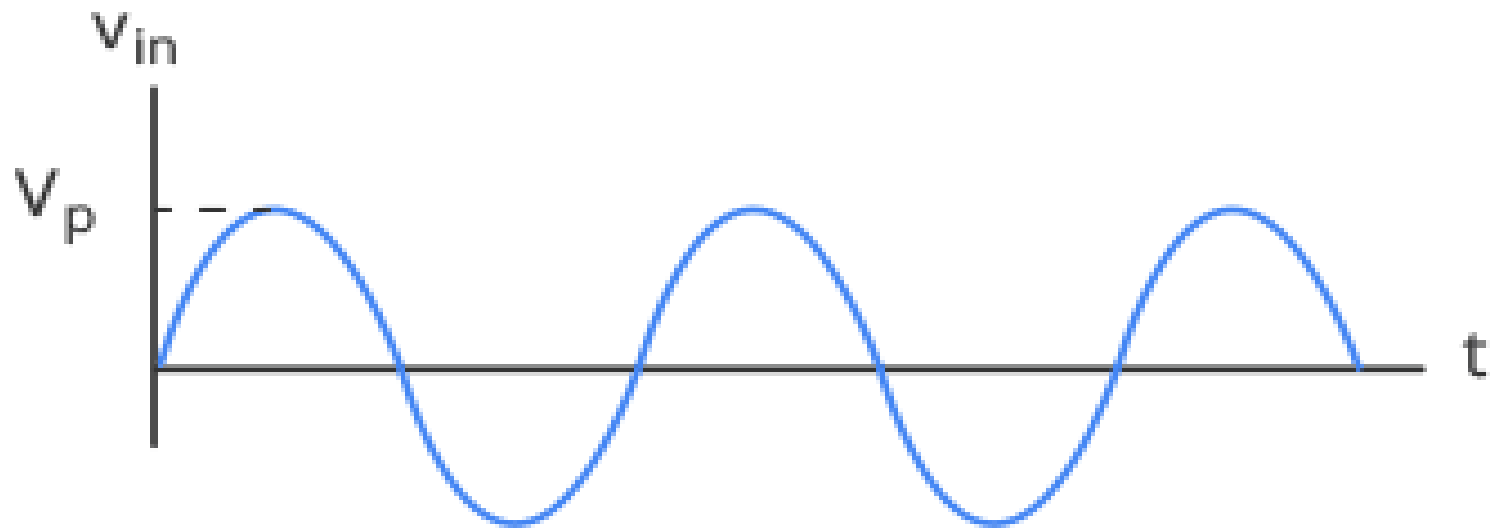
**Full Wave Bridge Rectifier Circuit Diagram and Waveform**



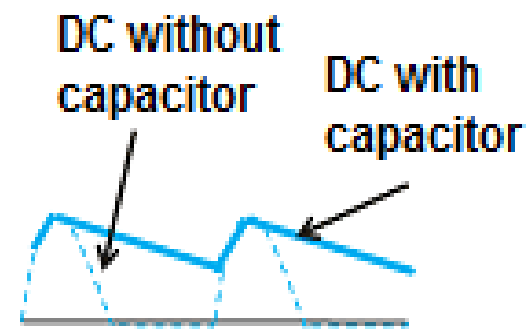
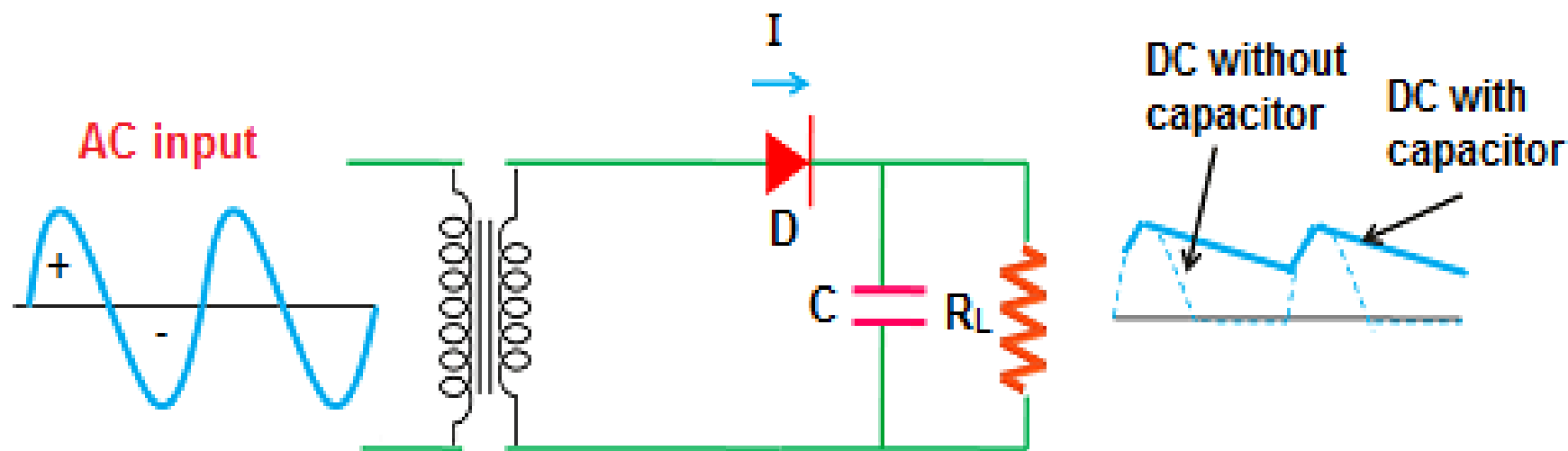
# Bridge Rectifier



# Bridge Rectifier Waveforms



# HW Rectifier With Capacitor Filter

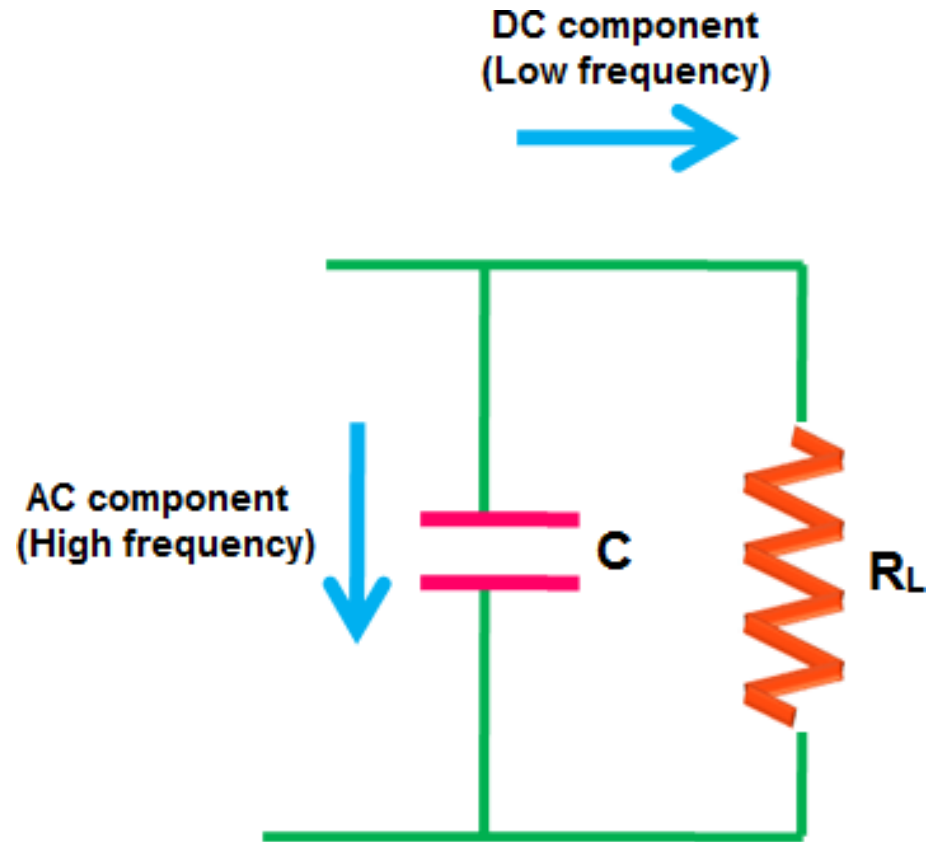


DC output

D = Diode  
C = Capacitor  
 $R_L$  = Load resistor  
I = Current

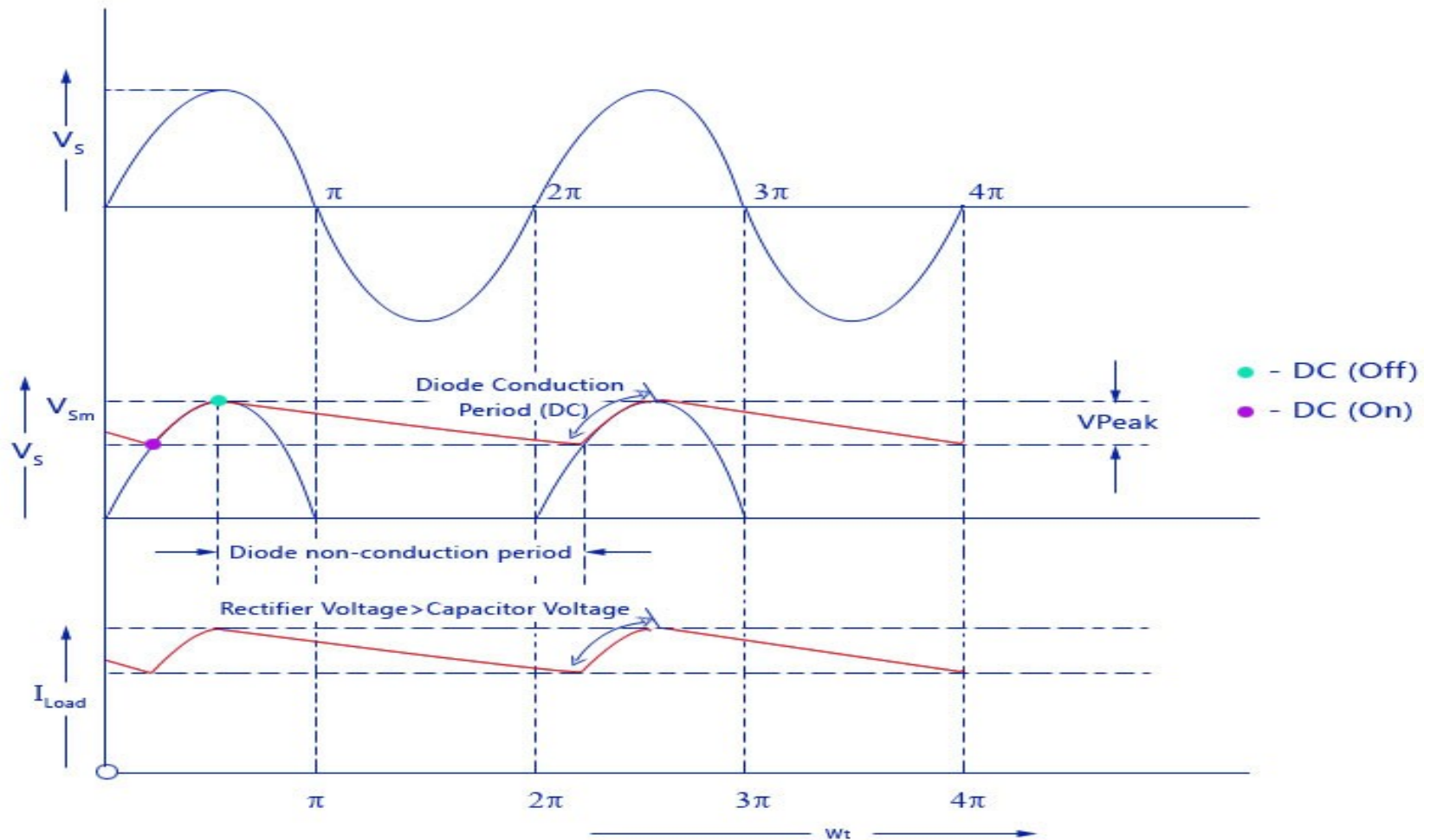
Half wave rectifier with capacitor filter

# HW Rectifier With Capacitor Filter

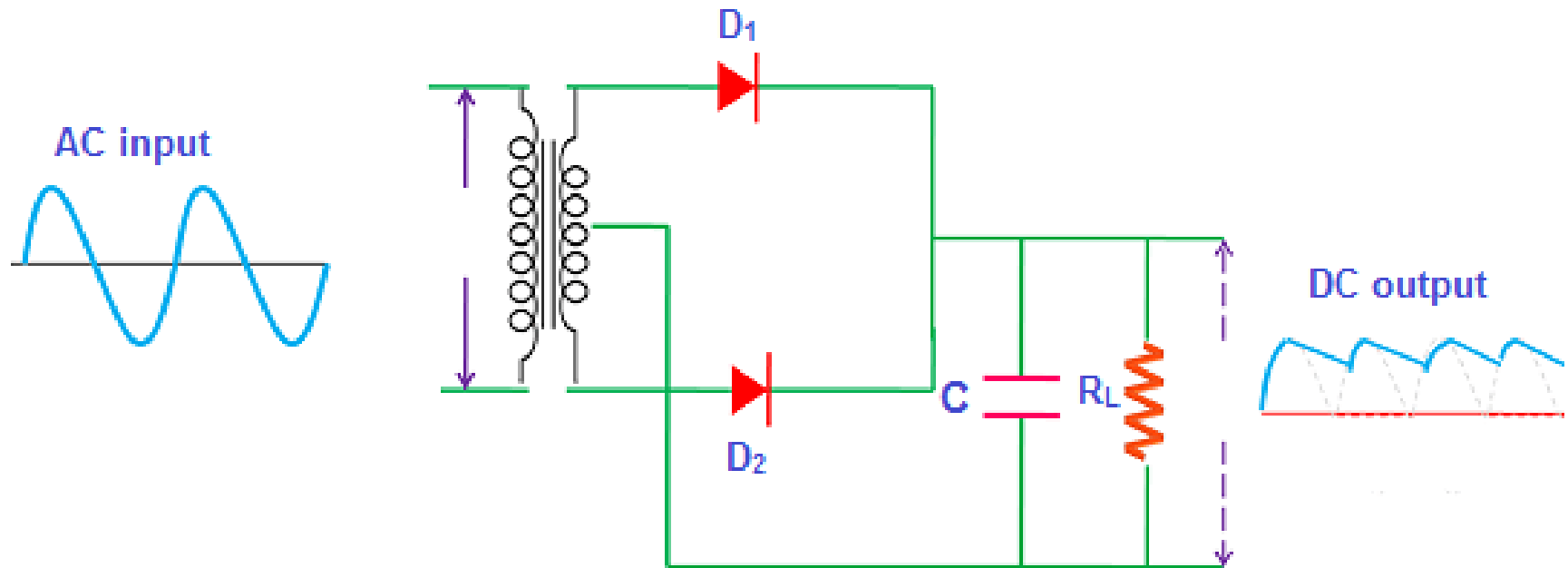


# Waveforms

## Half wave Rectifier with Capacitor Filter - Waveform



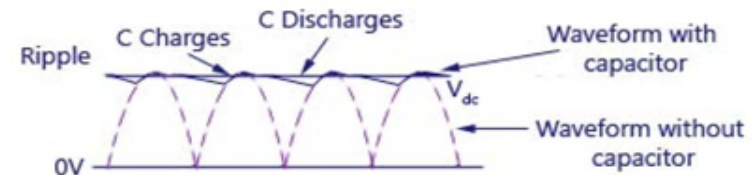
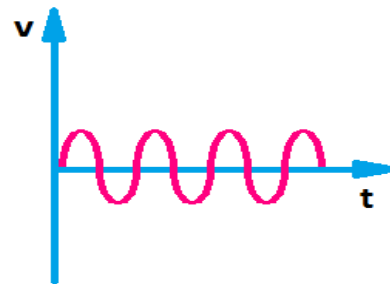
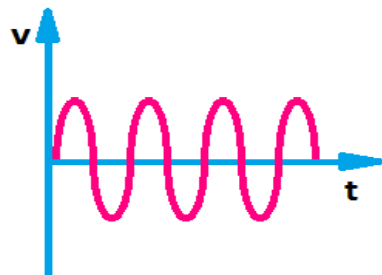
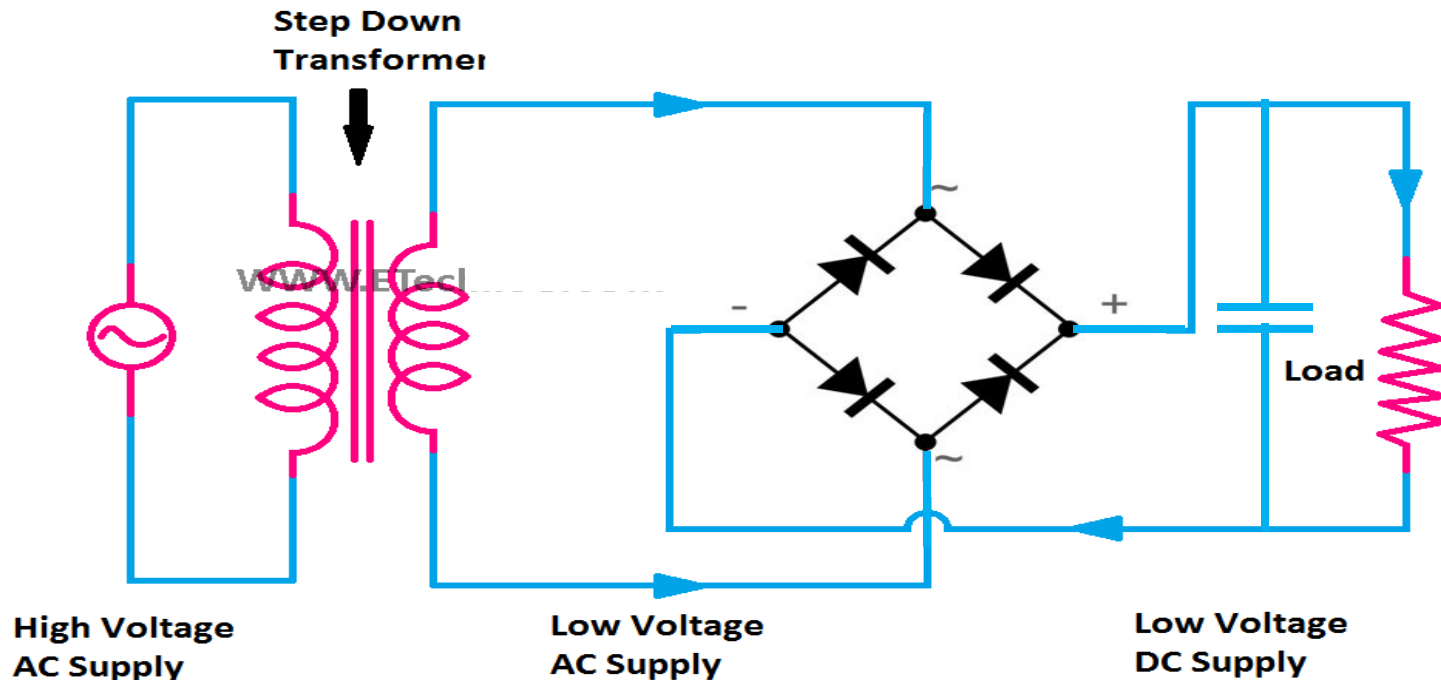
# CT FW Rectifier with Capacitor Filter



Full wave rectifier with capacitor filter

# Bridge Rectifier with Capacitor Filter

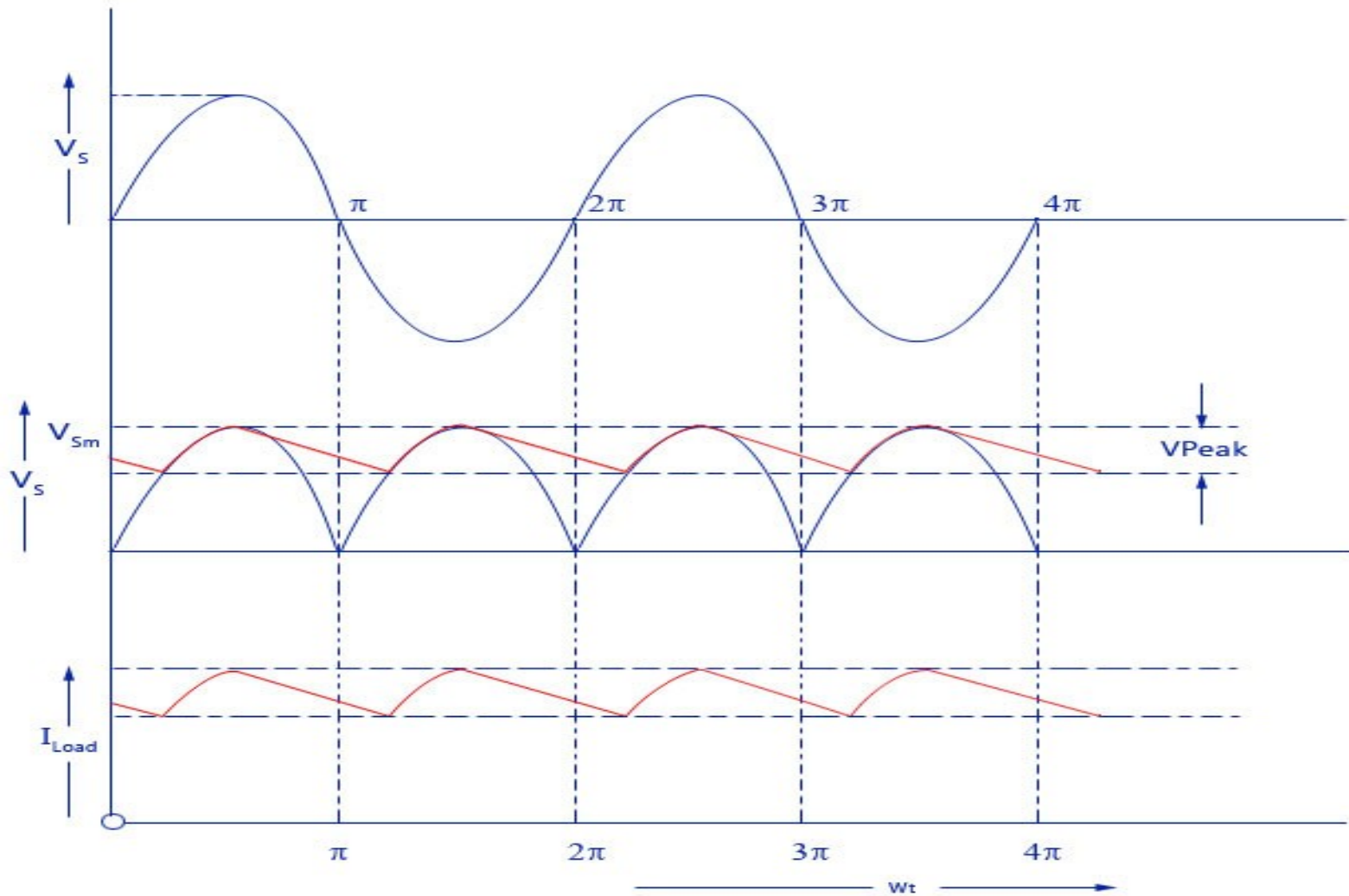
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**Full Wave Bridge Rectifier Circuit Diagram and Waveform**

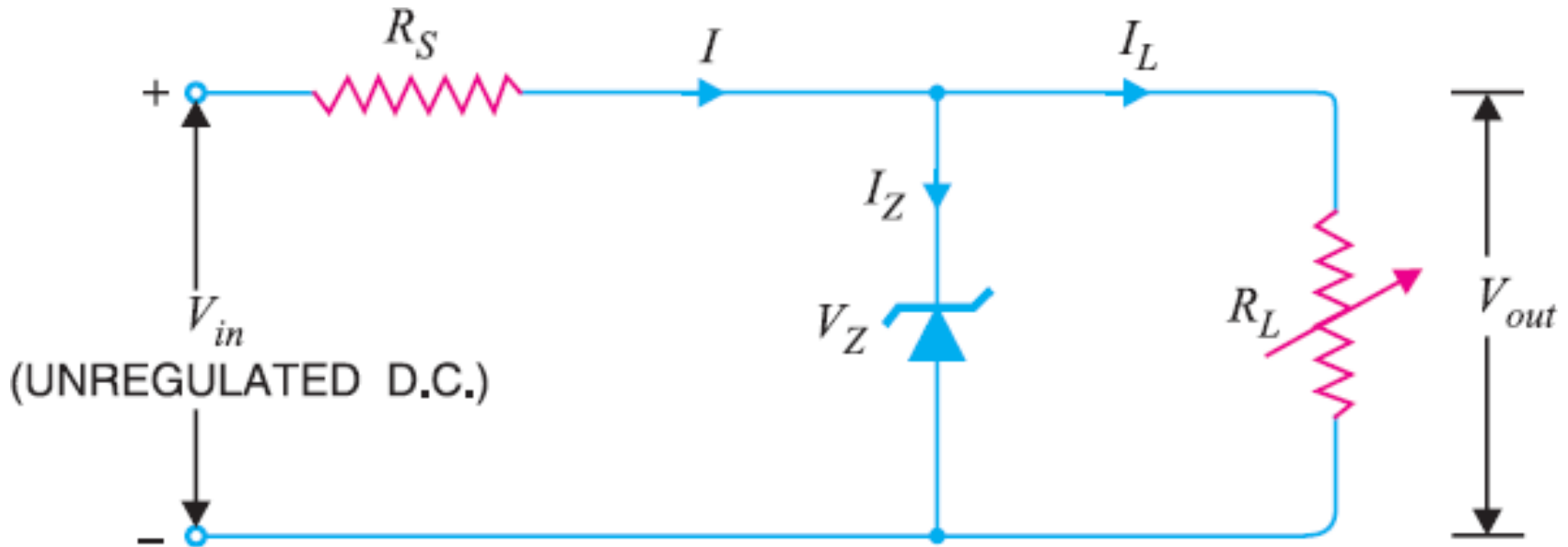
# Waveforms

## Fullwave Rectifier with Capacitor Filter - Waveform





# Zener Diode Regulator



When the field is about  $3 \times 10^7 \text{ V/m}$  electrons are pulled from the covalent bonds. A large number of electron holes pairs are thus produced and the reverse current sharply increases. This is known as the **zenar effect**.

# Function of Zener Diode

1.Zener diodes are a special kind of diode which permits current to flow in the forward direction.

2.Zener diodes will also allow current to flow in the reverse direction when the voltage is above a certain value. This breakdown voltage is known as the Zener voltage.

In a standard diode, the Zener voltage is high, and the diode is permanently damaged if a reverse current above that value is allowed to pass through it.

In the reverse bias direction, there is practically no reverse current flow until the breakdown voltage is reached.

When this occurs there is a sharp increase in reverse current.

Varying amount of reverse current can pass through the diode without damaging it.

The breakdown voltage or zener voltage ( $V_Z$ ) across the diode remains relatively constant.

# Zener Diode As A Voltage Regulator

A voltage regulator is an electronic circuit that provides a stable DC voltage independent of the load current, temperature and AC line voltage variations.

A Zener diode of break down voltage  $(V_Z)$  is reverse connected to an input voltage source  $(V_I)$  across a load resistance  $(R_L)$  and a series resistor  $(R_S)$ .

The voltage across the zener will remain steady at its break down voltage  $(V_Z)$  for all the values of zener current  $(I_Z)$  as long as the current remains in the break down region.

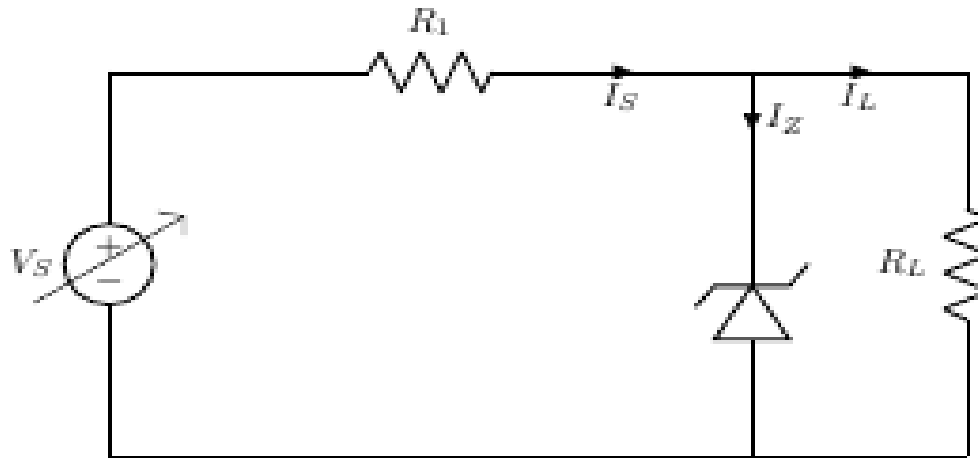
Hence a regulated DC output voltage  $(V_0 = V_Z)$  is obtained across  $(R_L)$ , whenever the input voltage remains within a minimum and maximum voltage.

Basically there are two type of regulations such as:

## Line Regulation:

In this type of regulation, series resistance and load resistance are fixed, only input voltage is changing.

Output voltage remains the same as long as the input voltage is maintained above a minimum value.



In Line Regulation, Load resistance is constant and input voltage varies.

$V_I$  must be sufficiently large to turn the Zener Diode ON.

$$V_L = V_Z = \frac{V_{Imin} \times R_L}{(R_S + R_L)}$$

So, the minimum turn-on voltage  $V_{Imin}$  is :

$$V_{Imin} = \frac{V_Z \times (R_S + R_L)}{R_L}$$

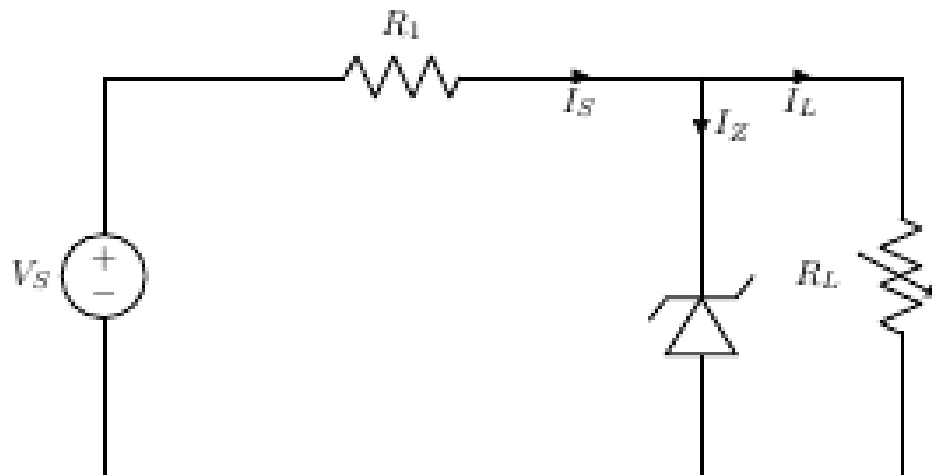
The maximum value of  $V_I$  is limited by the maximum zener current  $I_{Zmax}$

$$I_{Rmax} = I_{Zmax} + I_L$$

## Load Regulation:

In this type of regulation, input voltage is fixed and the load resistance is varying.

Output voltage remains same, as long as the load resistance is maintained above a minimum value.





In Load Regulation , input voltage is constant and Load resistance varies.  
Too small a Load Resistance  $R_L$  ,will result  $V_Z$  and Zener Diode will be OFF.

$$V_L = V_Z = \frac{V_{Imin} \times R_L}{(R_S + R_L)}$$

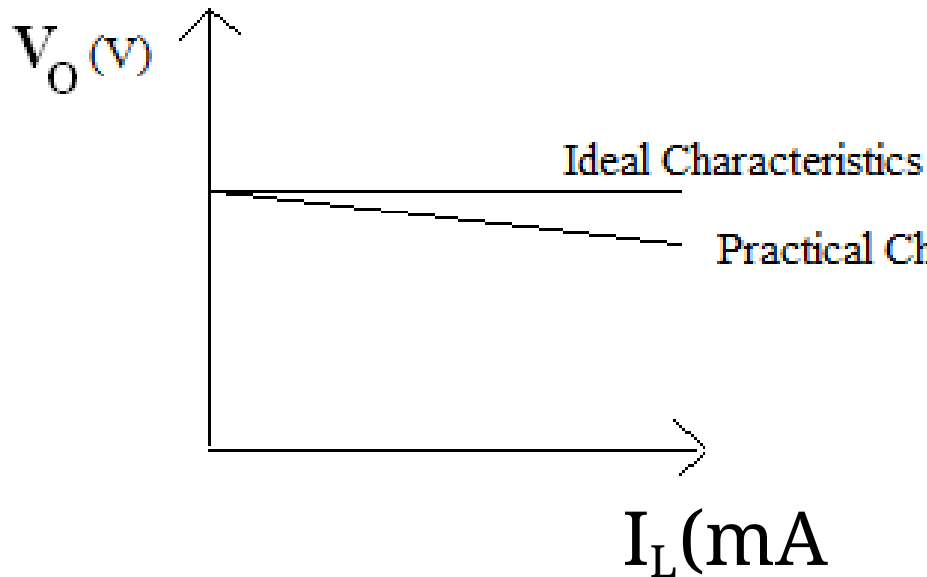
So the minimum load resistance  $R_L$

$$R_{Lmin} = \frac{V_Z \times R_S}{V_I - V_Z}$$

Any load resistance greater than  $R_{Lmin}$  will make Zener Diode ON

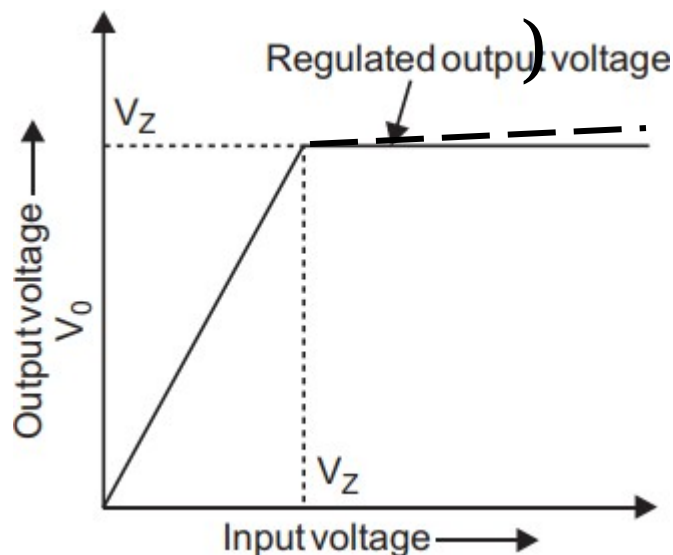
$$I_S = I_L + I_Z$$

# Zener Diode Regulator



Load regulation is the ability of a power supply output to remain constant given changes in the load.

**Load Regulation**



Line regulation is the ability of a power supply to maintain the output voltage given changes in the input line voltage

**Line Regulation**

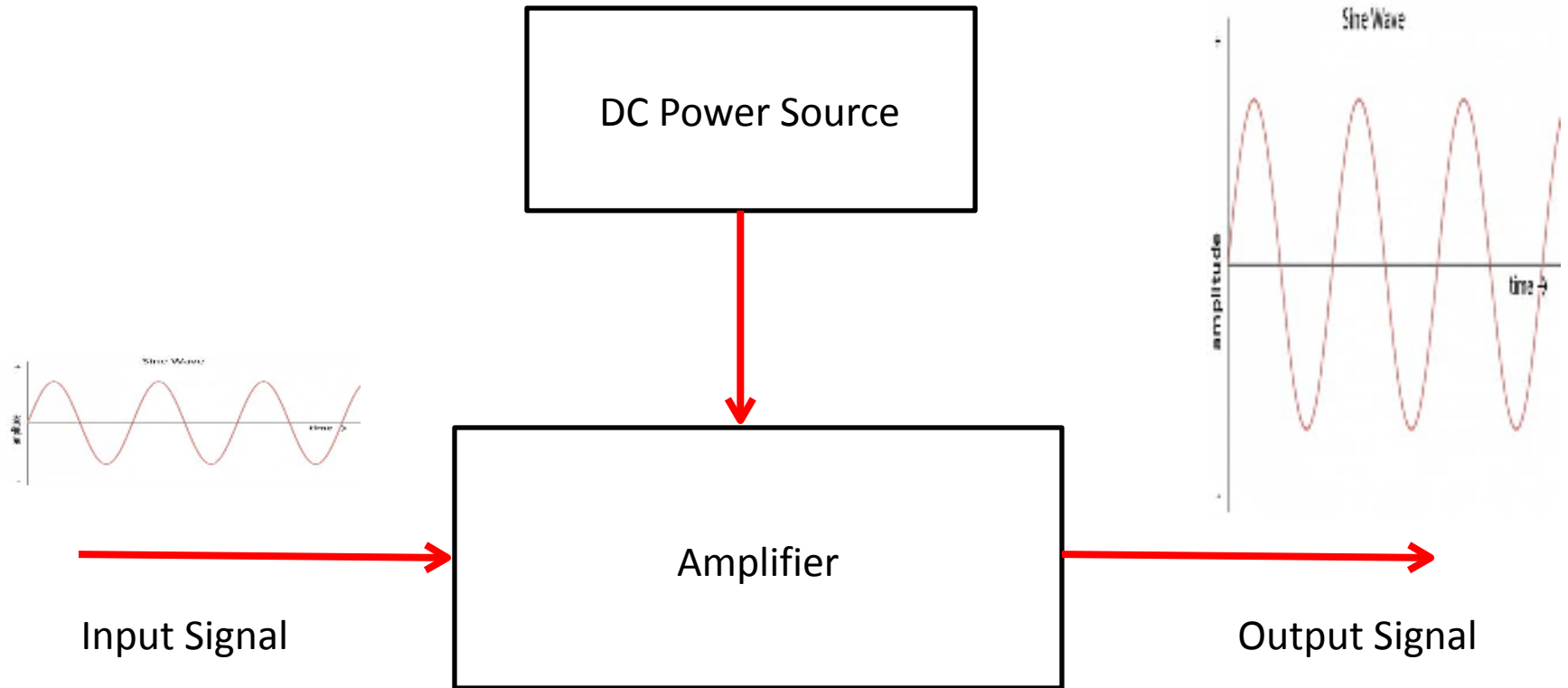
# Amplifiers

An **amplifier** is a device for increasing the power of a signal.

It does this by taking power from a power supply and controlling the output to match the input signal shape but with a larger amplitude.

An idealized amplifier can be said to be "a piece of wire with gain", as the output is an exact replica of the input, but larger.

# BD of Amplifier



# Characterestics of an Amplifier

- Input Resistance
- Output Resistance
- Voltage Gain
- Current Gain
- Power Gain

# Characteristics of an Amplifier

$$\text{Voltage Gain } A_v = \frac{V_{out}}{V_{in}}$$

$$\text{Current Gain } A_i = \frac{I_{out}}{I_{in}}$$

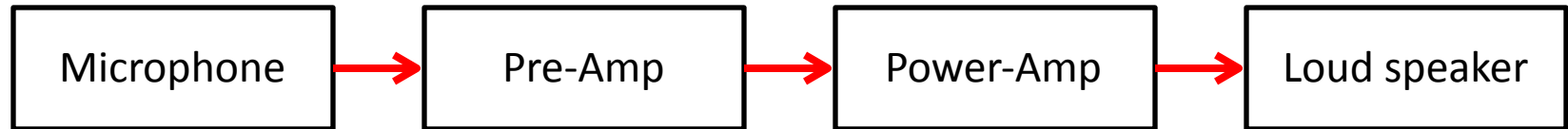
$$\text{Power Gain } A_p = \frac{P_{out}}{P_{in}}$$

$$\text{Power Gain } A_p \text{ (dB)} = 10 \log \frac{P_{out}}{P_{in}}$$

$$\text{Voltage Gain } A_v \text{ (dB)} = 20 \log \frac{V_{out}}{V_{in}}$$

$$\text{Current Gain } A_i \text{ (dB)} = 20 \log \frac{I_{out}}{I_{in}}$$

# Public Address system



# Hearing Range

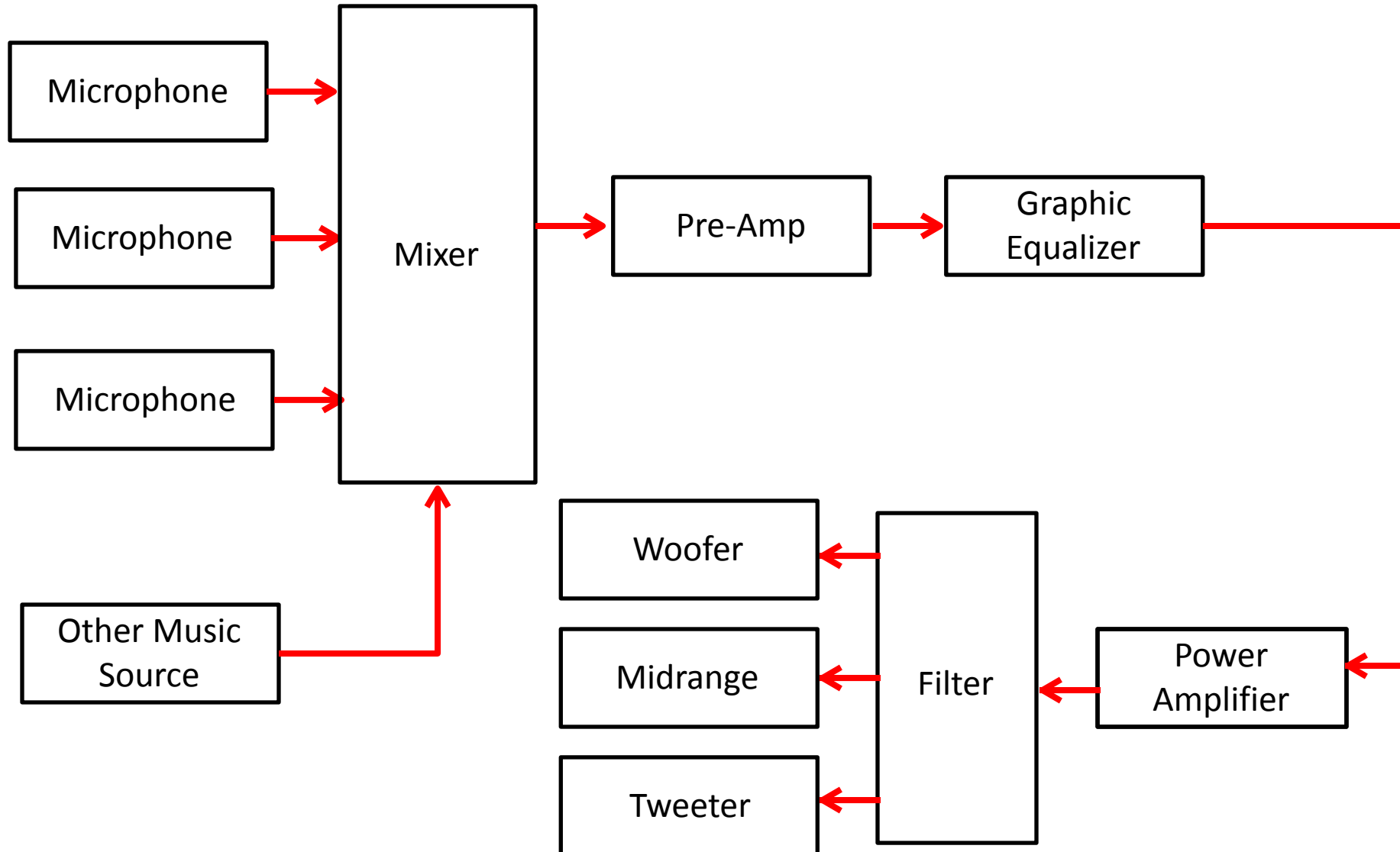
| Species  | Approximate Range (Hz) |
|----------|------------------------|
| Human    | 64-23,000              |
| Dog      | 67-45,000              |
| Cat      | 45-64,000              |
| Cow      | 23-35,000              |
| Rat      | 200-76,000             |
| Mouse    | 1,000-91,000           |
| Elephant | 16-12,000              |

Voice Frequency- Female:350Hz to 17 KHz

Male: 100Hz to 8 KHz



# Public Address system



# Speaker ranges

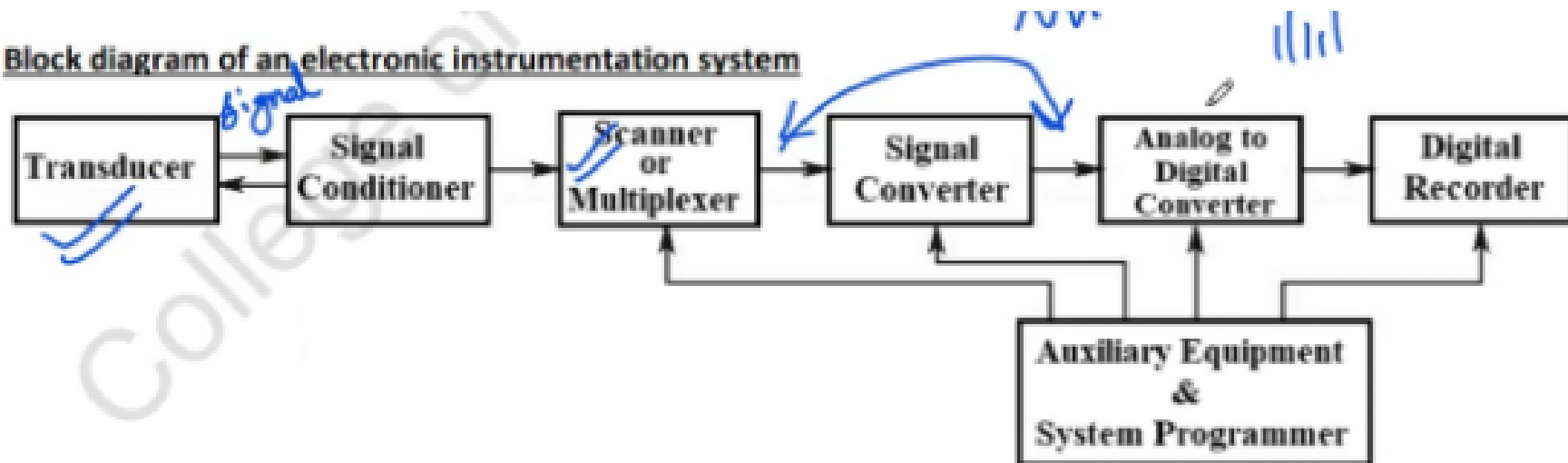
Standard Woofer: 20 Hz up to 2,000 Hz

Sub woofers: lower than 200 Hz

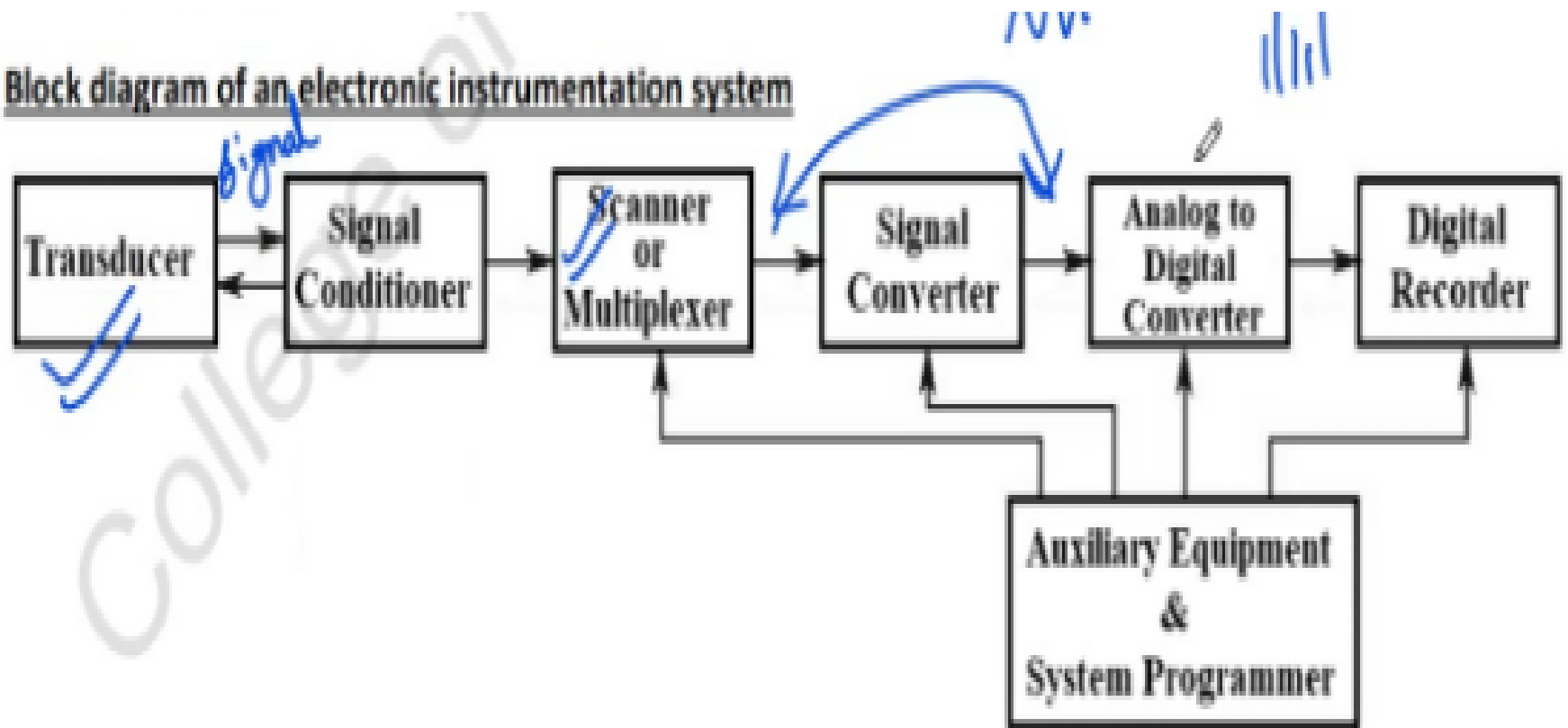
Mid range speakers: 500 Hz-4 kHz

Tweeters: 2 kHz-20kHz

Block diagram of an electronic instrumentation system

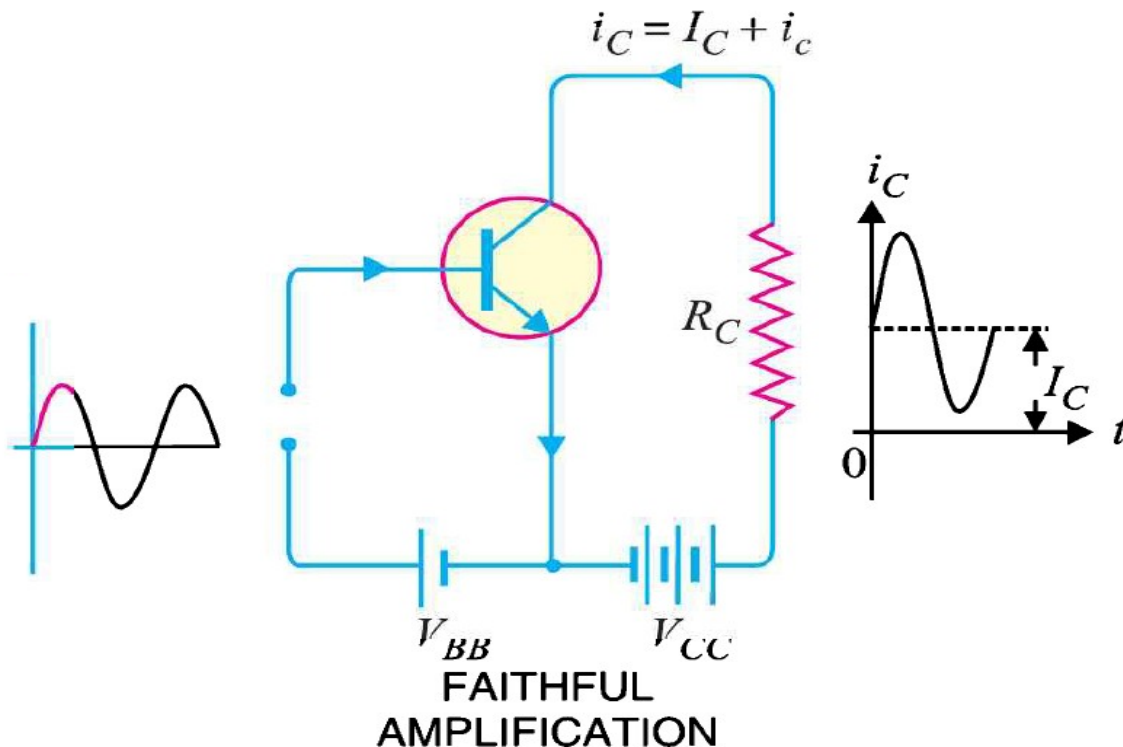


Block diagram of an electronic instrumentation system



# Need for biasing

**Ensure FR operation of Transistor**  
**Proper zero signal collector current.**  
**Proper minimum base-emitter voltage.**



# Biasing

Biasing is the process of providing DC voltage which helps in the functioning of the circuit.

A transistor is biased in order to make the emitter base junction forward biased and collector base junction reverse biased,

so that it maintains in active region, to work as an amplifier.

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is known as **Transistor Biasing**.

The circuit which provides transistor biasing is called as **Biasing Circuit**.

## Need for DC biasing

If a signal of very small voltage is given to the input of BJT, it cannot be amplified.

Because, for a BJT, to amplify a signal, two conditions have to be met.

The input voltage should exceed **cut-in voltage** for the transistor to be **ON**.

The BJT should be in the **active region**, to be operated as an **amplifier**

If appropriate DC voltages and currents are given through BJT by external sources,

so that BJT operates in active region and superimpose the AC signals to be amplified,

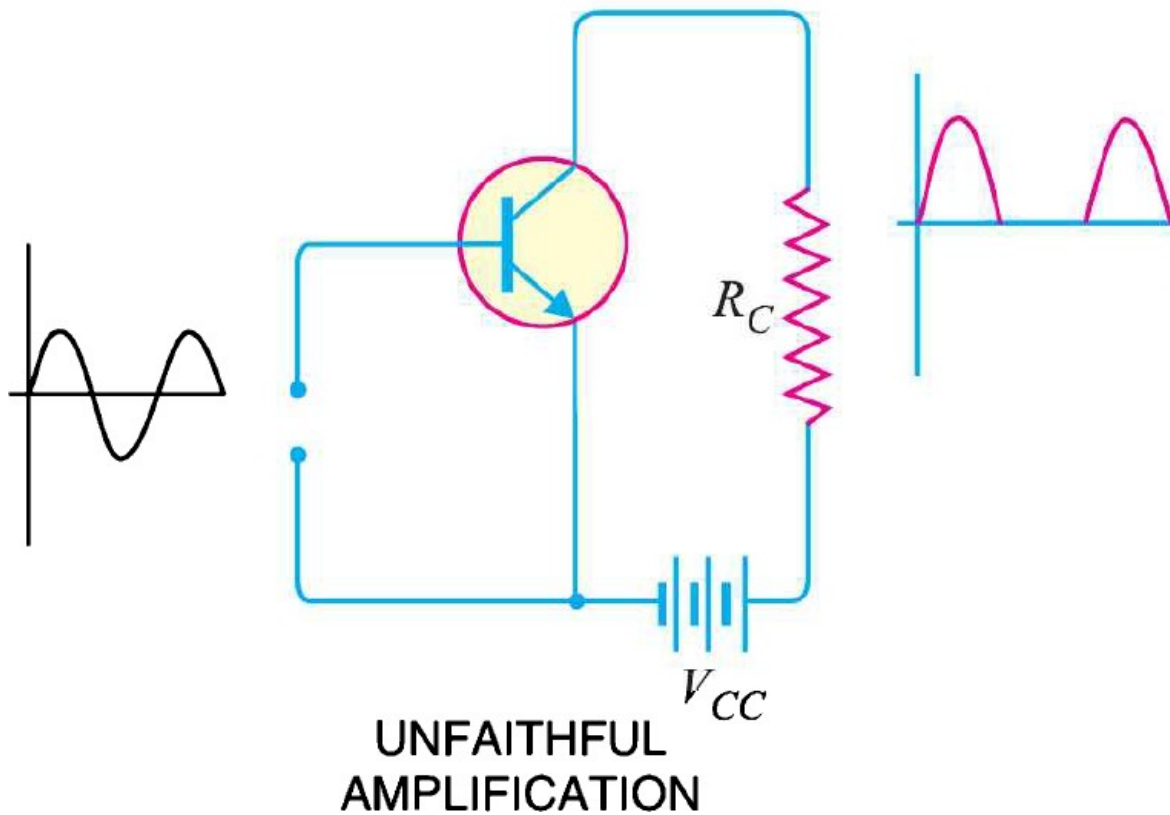
then this problem can be avoided.

The given DC voltage and currents are so chosen that the transistor remains in active region for entire input AC cycle.

Hence DC biasing is needed.

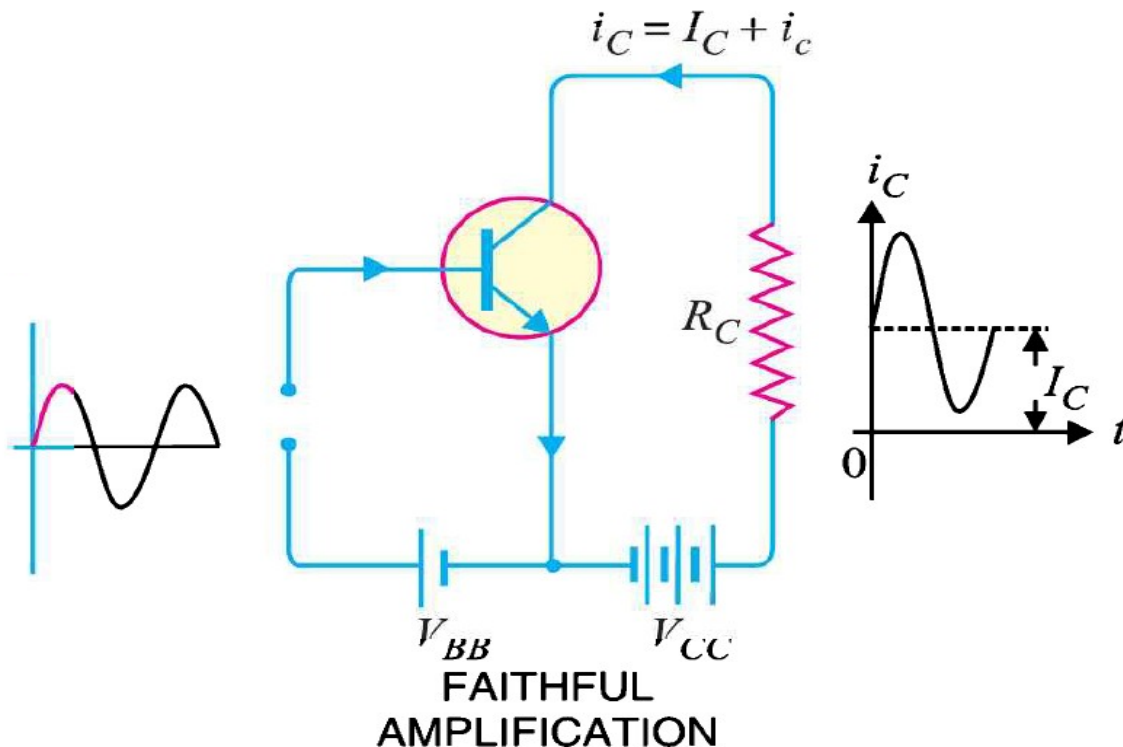


# Need for biasing



# Need for biasing

**Ensure FR operation of Transistor**  
**Proper zero signal collector current.**  
**Proper minimum base-emitter voltage.**



# Need for biasing

Beta and  $V_{BE}$  varies with temperature

*The process of making operating point independent of temperature changes or variations in transistor parameters is known as **stabilisation**.*

**Need for stabilisation.** Stabilisation of the operating point is necessary due to the following reasons :

- (i) Temperature dependence of  $I_C$
- (ii) Individual variations
- (iii) Thermal runaway

# Voltage divider circuit

The DC voltage is called the bias voltage and its magnitude is such as to keep the input circuit is always forward biased regardless the polarity of the input signal

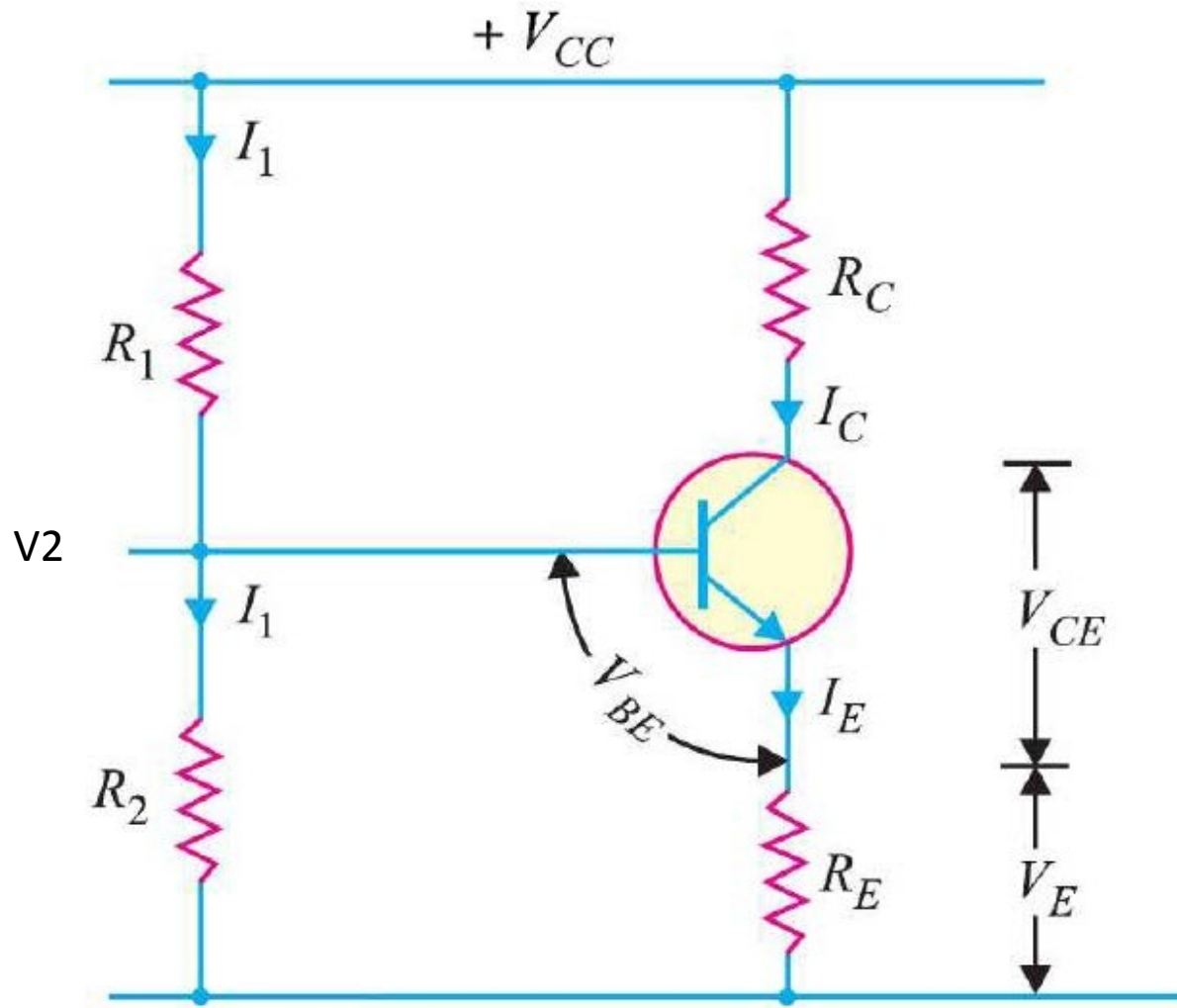
In this circuit R1 and R2 (forms the voltage divider) are connected across Vcc.

$R_E$  connected for stability.

The voltage drop across R2 f/b BE junction and causes  $I_B$ .

Q point is independent of  $\beta$ .

# Voltage divider circuit



# Analysis

**Analysis can be done by approximation or accurate method thevenin theorem.**

**Consider input section, Let current  $I_1$  flows through  $R_1$ , As  $I_B$  is very small assume  $I_1$  is flowing through  $R_2$ ,**

**Applying KVL**

$$V_{CC} = I_1 R_1 + I_1 R_2$$

**Then**

# Voltage divider circuit

$$I_1 = \frac{V_{CC}}{R_1 + R_2}$$

Voltage across resistance  $R_2$  is

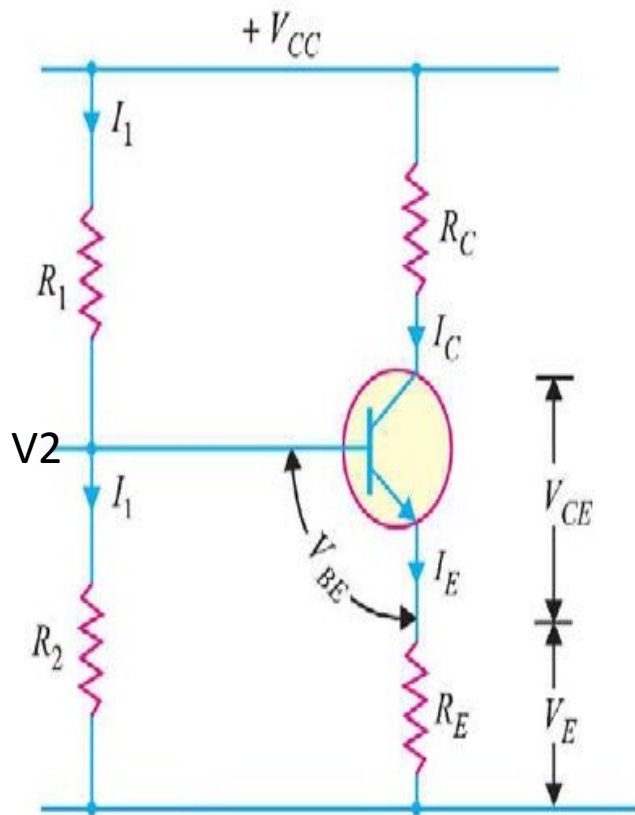
$$V_2 = \left( \frac{V_{CC}}{R_1 + R_2} \right) R_2$$

Applying KVL to base emitter loop

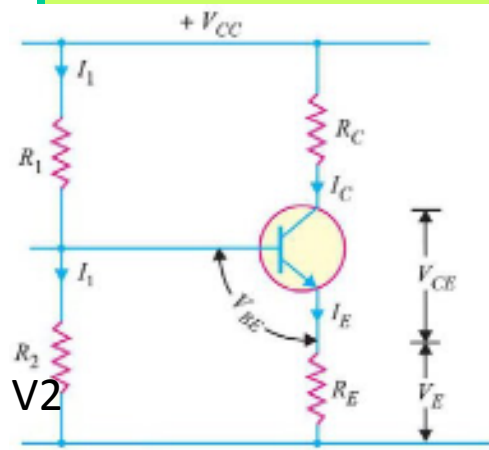
$$V_2 = V_{BE} + V_E$$

$$V_2 = V_{BE} + I_E R_E$$

$$I_E = \frac{V_2 - V_{BE}}{R_E}$$



# voltage divider circuit



Applying KVL to  
collector emitter (o/p)  
loop

$$\begin{aligned} V_{CC} &= I_C R_C + V_{CE} + I_E R_E \\ &= I_C R_C + V_{CE} + I_C R_E \\ &= I_C (R_C + R_E) + V_{CE} \end{aligned}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Since  $I_E \approx I_C$

(i) Collector current  $I_C$  :

$$I_C = \frac{V_2 - V_{BE}}{R_E}$$

*$I_C$  Does not  
depend on  
beta*

(1)

OPERATING  
POINT

(2)



**From the equations (1) and (2)  $I_C$  and  $V_{CE}$  can be determined and Q point is established.**

**It is clear from the equation (1) that  $I_C$  does not depends on  $\beta$ .**

**Depends on  $V_{BE}$  but it is very small.**

**So  $I_C$  is independent of transistor parameters, hence good stabilization is ensured.**

# Factors affecting the operating point

The main factor that affect the operating point is the temperature.

The operating point shifts due to change in temperature.

As temperature increases, the values of  $I_{CE}$ ,  $\beta$ ,  $V_{BE}$  gets affected.

$I_{CBO}$  gets doubled (for every  $10^\circ$  rise)

$V_{BE}$  decreases by 2.5mv (for every  $1^\circ$  rise)

So the main problem which affects the operating point is temperature.

Hence operating point should be made independent of the temperature so as to achieve stability.

To achieve this, biasing circuits are introduced.

# Stabilization

The process of making the operating point independent of temperature changes or variations in transistor parameters is known as **Stabilization**.

Once the stabilization is achieved, the values of  $I_C$  and  $V_{CE}$  become independent of temperature variations or replacement of transistor.

A good biasing circuit helps in the stabilization of operating point.

# Need for Stabilization

Stabilization of the operating point has to be achieved due to the following reasons.

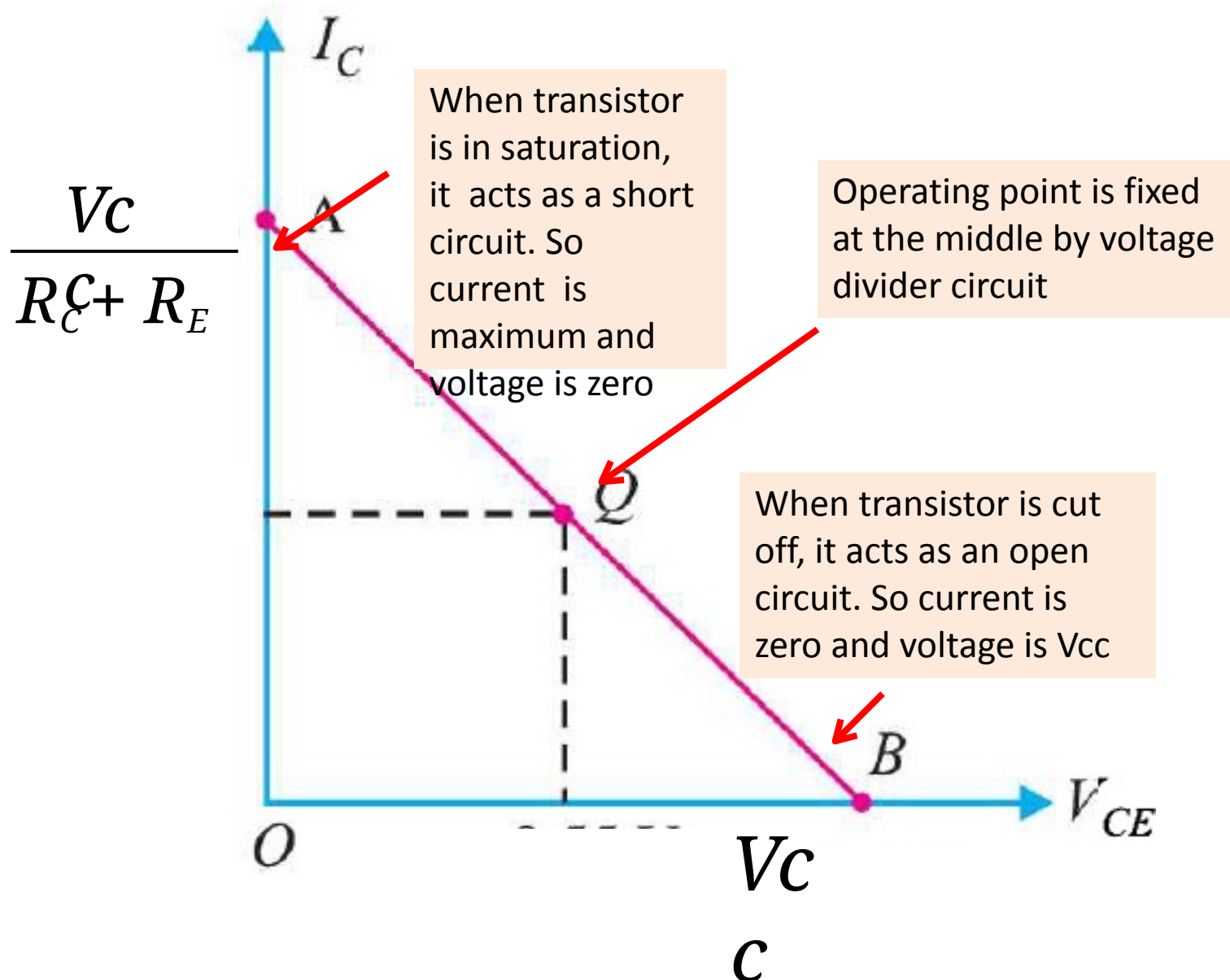
Temperature dependence of  $I_C$

Individual variations

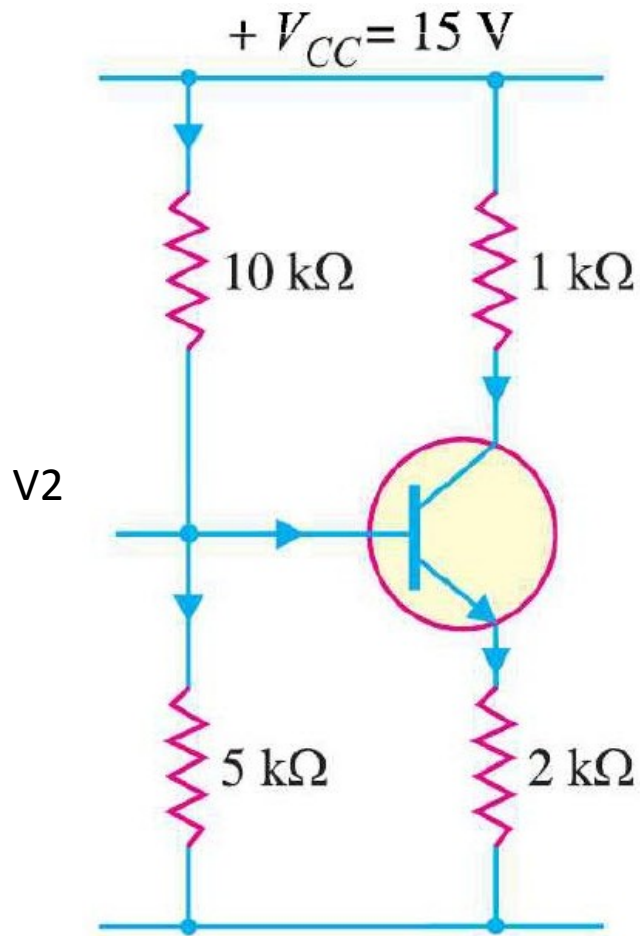
Thermal runaway

Let us understand these concepts in detail.

# Voltage divider circuit-DC Load Line



# Voltage divider circuit

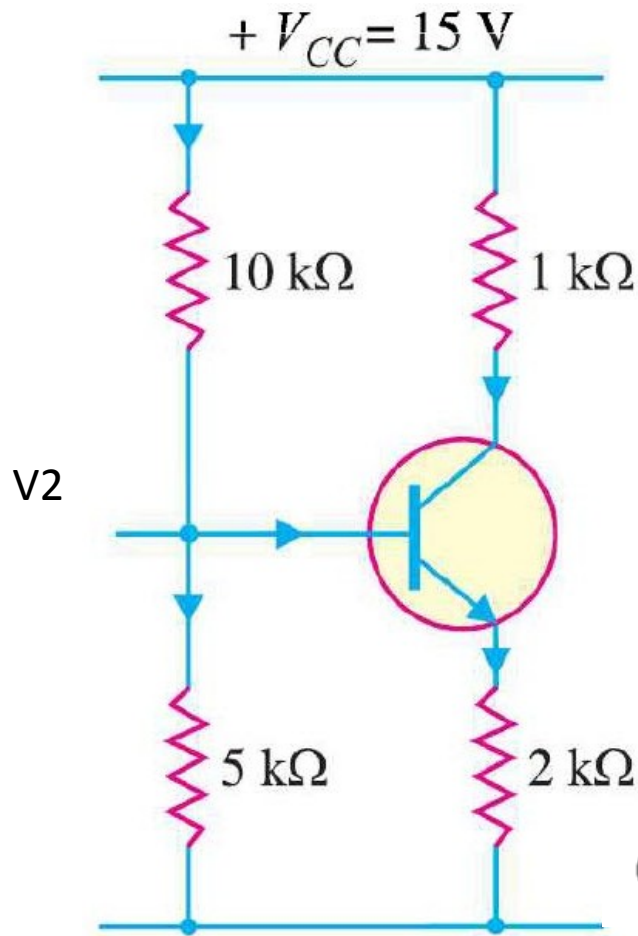


$$V_{BE} = 0.7 \text{ V}$$

$$V_2 = \frac{V_{CC}}{10 + 5} \times 5$$
$$= \frac{15 \times 5}{10 + 5} = 5 \text{ V}$$

$$I_E = \frac{V_2 - V_{BE}}{R_E}$$
$$= \frac{5 - 0.7}{2 \text{ k}\Omega} = \frac{4.3 \text{ V}}{2 \text{ k}\Omega} = 2.15 \text{ mA}$$

# Voltage divider circuit



$$I_C \simeq I_E = 2.15 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

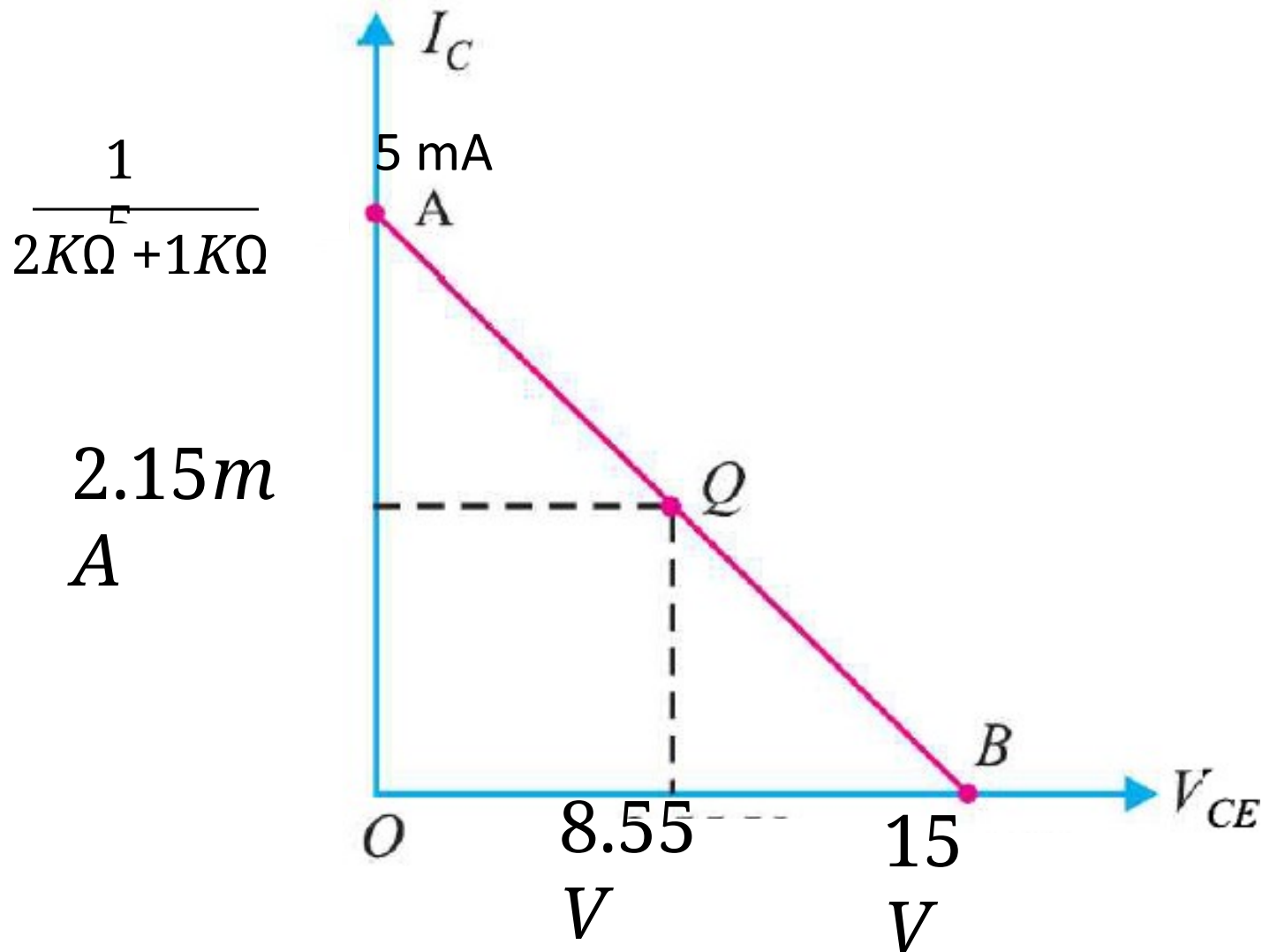
$$= 15 - 2.15 \text{ mA} \times 3 \text{ k}\Omega$$

$$= 15 - 6.45 = 8.55 \text{ V}$$

Operating point is **8.55 V, 2.15 mA**.



# DC Load Line



# RC coupled Amplifier

*Name of coupling*

RC coupling

Transformer coupling

Direct coupling

*Name of multistage amplifier*

R-C coupled amplifier

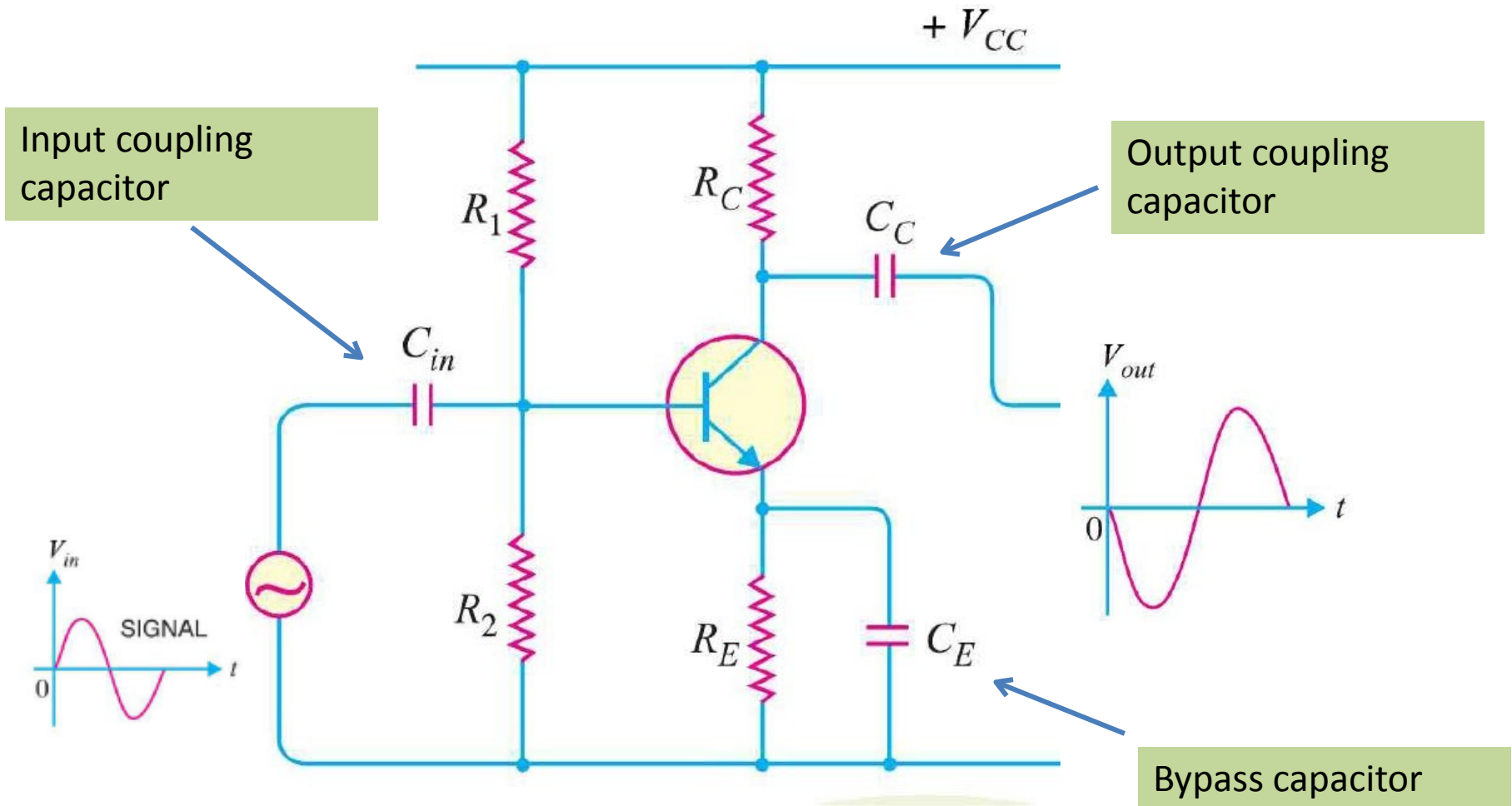
Transformer coupled amplifier

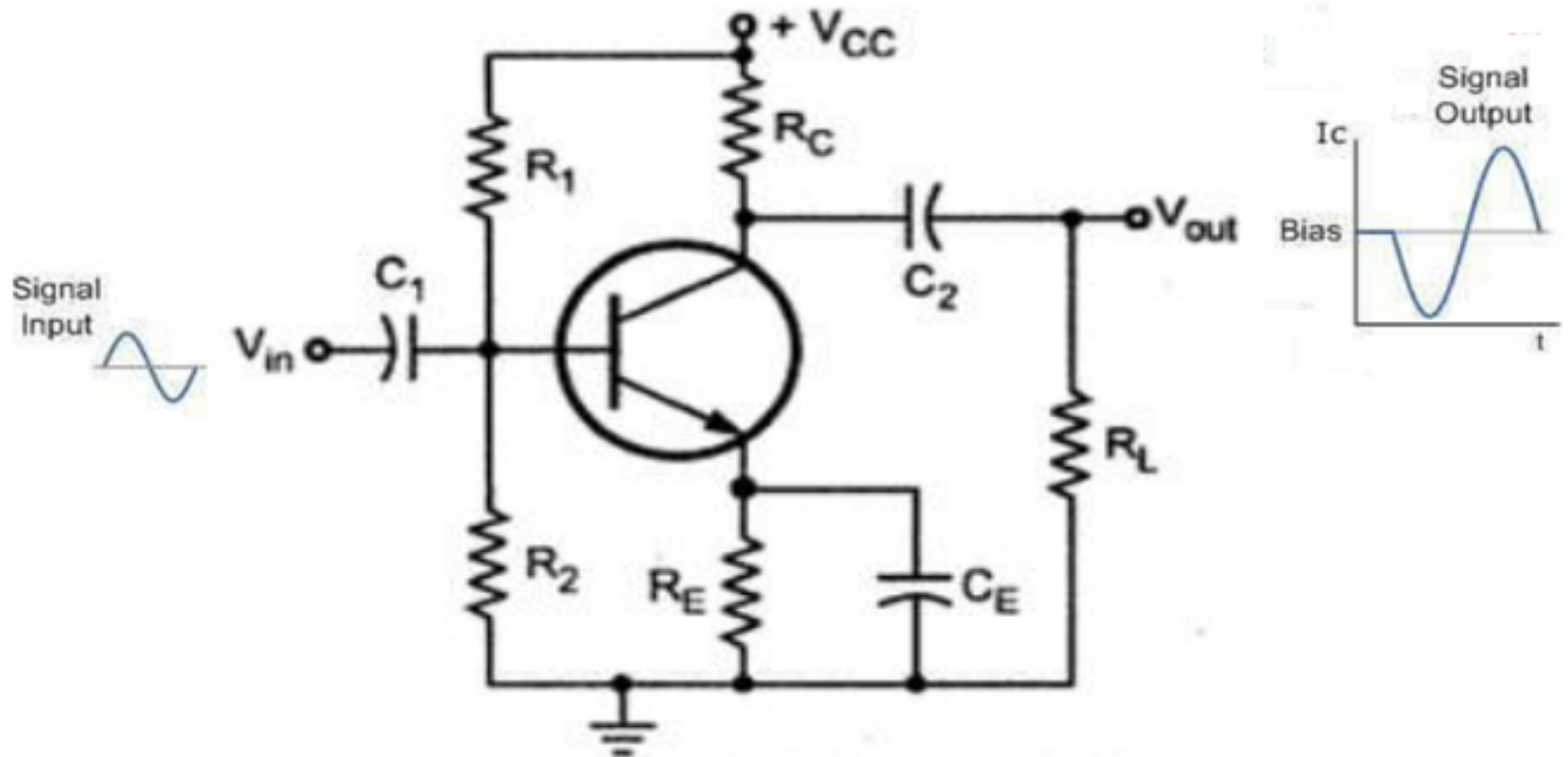
Direct coupled amplifier



(i) In *RC* coupling, a capacitor is used as the coupling device. The capacitor connects the output of one stage to the input of the next stage in order to pass the a.c. signal on while blocking the d.c. bias voltages.

# RC coupled Amplifier





**Typical RC coupled common emitter amplifier**

- Transistor acts as an amplifier when it is in active mode.
- i.e., Emitter-Base Junction (EBJ) is forward biased and Collector-Base Junction (CBJ) is reverse biased.
- Common Emitter (CE) amplifier is the widely used configuration because of its high voltage gain.

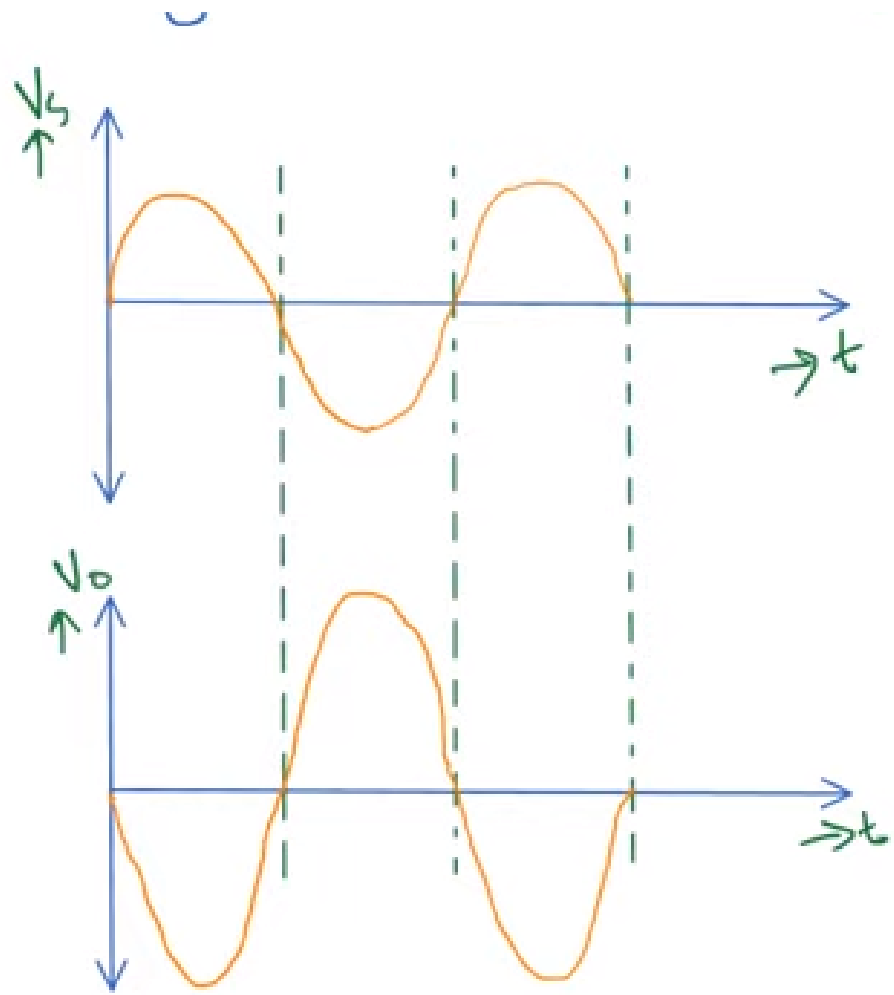
## Working

- In CE amplifier the input is applied between the Base and the Emitter.
- The output is taken from the Collector and the common Emitter terminals.
- In the first figure, the DC source  $V_{BB}$  and  $V_{CC}$  are used to bias the junctions of the transistor.

- In the second figure, we are using a Voltage Divider Biasing (VDB) Technique to bias the junctions of the transistor.
- $V_i$  is the input signal.
- The input is applied to the base of the transistor through capacitor  $C_1$  (Coupling capacitor).
- Coupling capacitor removes any DC components
- The output is taken through the capacitor  $C_2$ .  
The effect of DC at output will be removed by it.

- After biasing the transistor, i/p signal is applied.
- This produces fluctuations in the base current and it will be reflected in the collector current
- The output will be the enlarged version of the input having the same frequency but with  $180^\circ$  phase shift.





## Reason for $180^\circ$ phase shift

- Output voltage of CE amplifier  $V_o = V_{ce} = V_{cc} - I_c R_c$
- When the i/p voltage increases base current  $I_B$  increases.
  - It increases  $I_c$  ( $I_c = \beta I_B$ )
- When  $I_c$  increased, drop across  $R_c$  ( $I_c R_c$ ) increases.
- This reduces the  $V_o$ .

→ It increases  $I_C$  ✓ ( $I_C = \beta I_B$ )

$$\beta = I_C / I_B$$

→ When  $I_C$  increased, drop across  $R_C$  ( $I_C R_C$ ) increases. ↑

→ This reduces the  $V_o$ . ↓

$$V_s \downarrow \rightarrow I_B \downarrow \rightarrow I_C \downarrow \rightarrow I_C R_C \downarrow$$

→ When i/p voltage decreases the exact opposite will happen and the o/p voltage  $V_o$  increases.

$$\downarrow \\ V_o \uparrow$$

→ So this is the reason for  $180^\circ$  phase shift in CE amplifier.

# Operation of RC Coupled Amplifier

When an AC input signal is applied to the base of first transistor, it gets amplified and appears at the collector load  $R_L$  which is then passed through the coupling capacitor  $C_c$  to the next stage.

This becomes the input of the next stage, whose amplified output again appears across its collector load.

Thus the signal is amplified in stage by stage action.

The important point that has to be noted here is that the total gain is less than the product of the gains of individual stages.

This is because when a second stage is made to follow the first stage, the **effective load resistance** of the first stage is reduced due to the shunting effect of the input resistance of the second stage.

Hence, in a multistage amplifier, only the gain of the last stage remains unchanged.

As we consider a two stage amplifier here, the output phase is same as input. Because the phase reversal is done two times by the two stage CE configured amplifier circuit.

# Frequency Response of RC Coupled Amplifier

Frequency response curve is a graph that indicates the relationship between voltage gain and function of frequency. The frequency response of a RC coupled amplifier is as shown in the following graph.

From the above graph, it is understood that the frequency rolls off or decreases for the frequencies below 50Hz and for the frequencies above 20 KHz. whereas the voltage gain for the range of frequencies between 50Hz and 20 KHz is constant.

We know that,

Capacitive reactance (symbol  $X_C$ ) is a measure of a capacitor's opposition to AC (alternating current).

$$X_C = \frac{1}{2\pi f_c}$$

It means that the capacitive reactance is inversely proportional to the frequency

## At Low frequencies (i.e. below 50 Hz)

The capacitive reactance is inversely proportional to the frequency.

At low frequencies, the reactance is quite high. The reactance of input capacitor  $C_{in}$  and the coupling capacitor  $C_c$  are so high that only small part of the signal will pass from the amplifier stage to the load.

The reactance of the emitter by pass capacitor  $C_E$  is also very high during low frequencies.

Hence it cannot shunt the emitter resistance effectively. With all these factors, the voltage gain rolls off at low frequencies.



# At High frequencies (i.e. above 20 KHz)

Again considering the same point, we know that the capacitive reactance is low at high frequencies.

So, The reactance of coupling capacitor C2 is very small and it behaves as a short circuit.

As a result of this, the loading effect of the next stage increases,

which reduces the voltage gain.

Along with this, as the capacitance of emitter diode decreases, it increases the base current of the transistor due to which the current gain ( $\beta$ ) reduces. Hence the voltage gain rolls off at high frequencies.

# At Mid-frequencies (i.e. 50 Hz to 20 KHz)

The voltage gain of the capacitors is maintained constant in this range of frequencies, as shown in figure.

If the frequency increases, the reactance of the capacitor  $C_c$  decreases which tends to increase the gain.

But this lower capacitive reactance increases the loading effect of the next stage by which there is a reduction in gain.

Due to these two factors, the gain is maintained constant.

# Advantages of RC Coupled Amplifier

The following are the advantages of RC coupled amplifier.

The frequency response of RC amplifier provides constant gain over a wide frequency range, hence most suitable for audio applications.

The circuit is simple and has lower cost because it employs resistors and capacitors which are cheap. It becomes more compact with the upgrading technology.

# Disadvantages of RC Coupled Amplifier

The following are the disadvantages of RC coupled amplifier.

The voltage and power gain are low because of the effective load resistance.

They become noisy with age.

Due to poor impedance matching, power transfer will be low.

# Applications of RC Coupled Amplifier

The following are the applications of RC coupled amplifier.

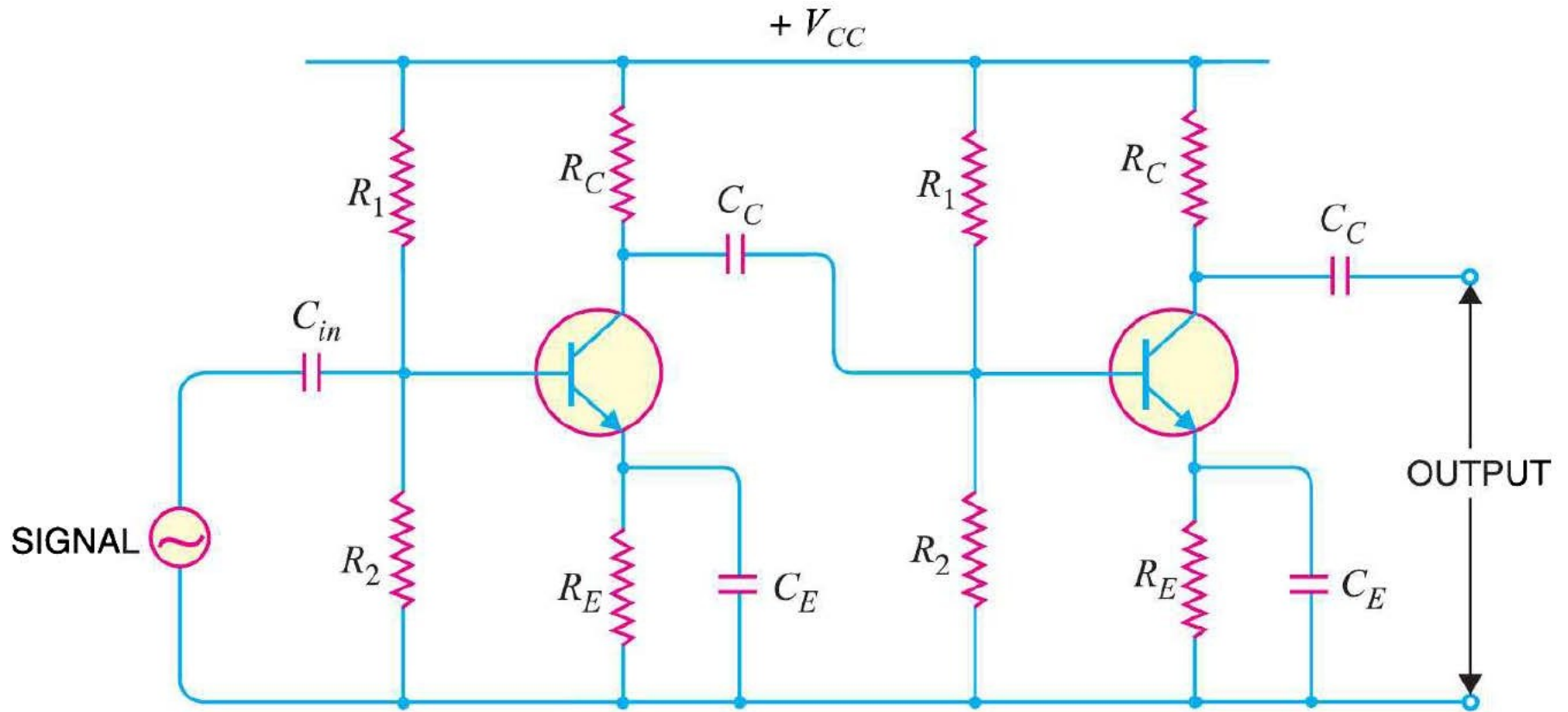
They have excellent audio fidelity over a wide range of frequency.

Widely used as Voltage amplifiers

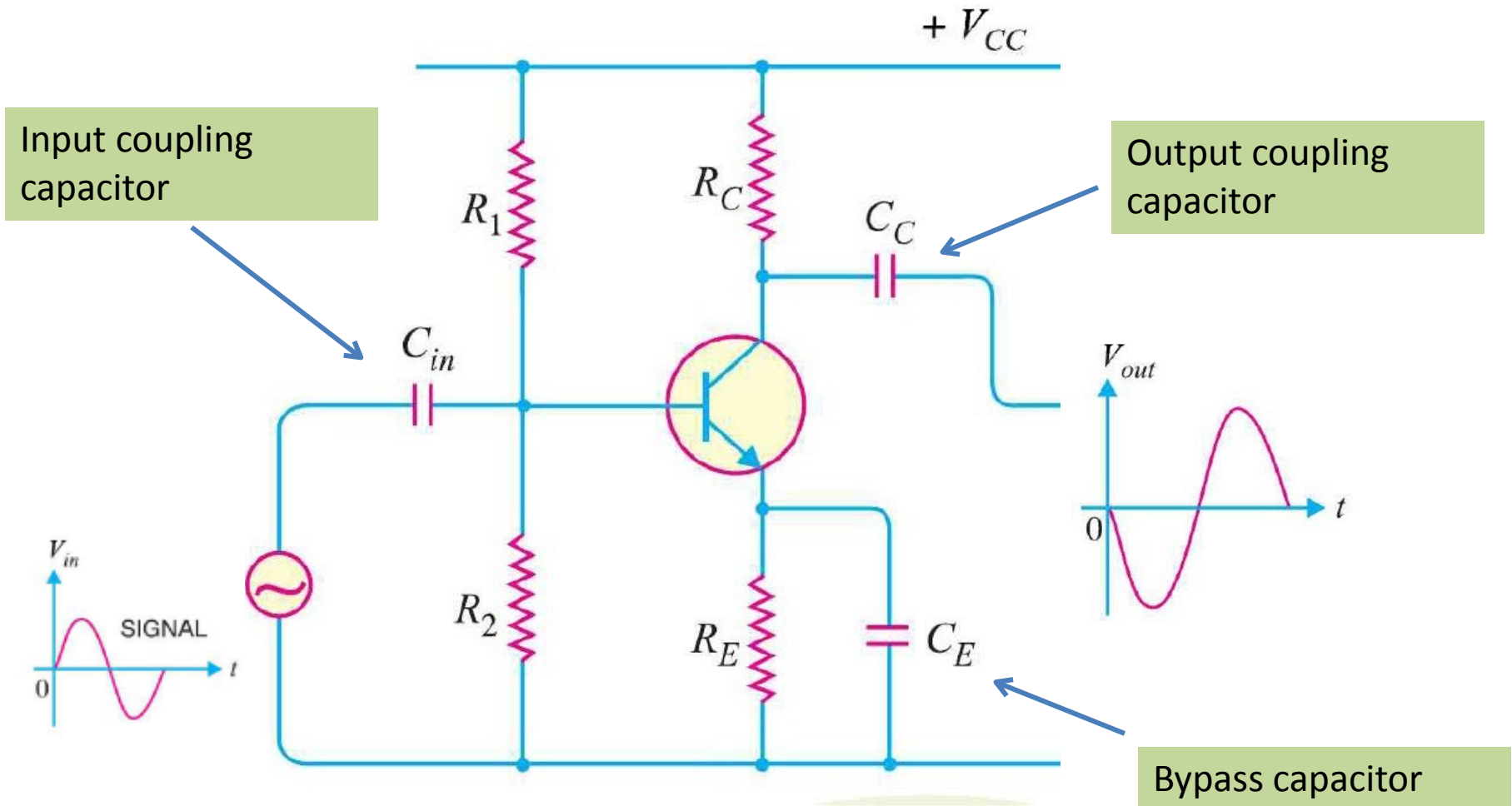
Due to poor impedance matching, RC coupling is rarely used in the final stages.

<https://www.youtube.com/watch?v=E-MH1sIA72E>

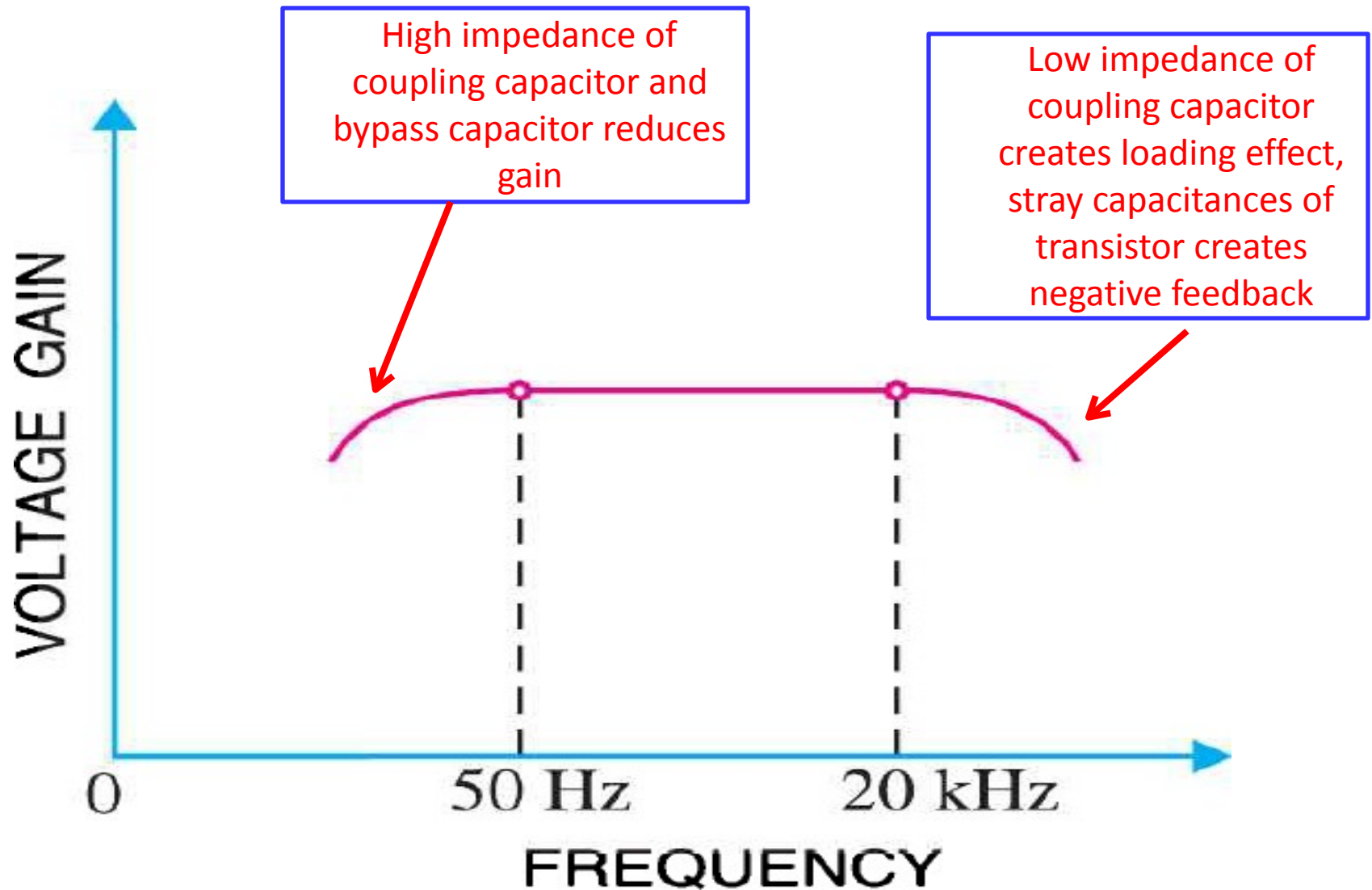
# RC coupled Amplifier



# RC coupled Amplifier



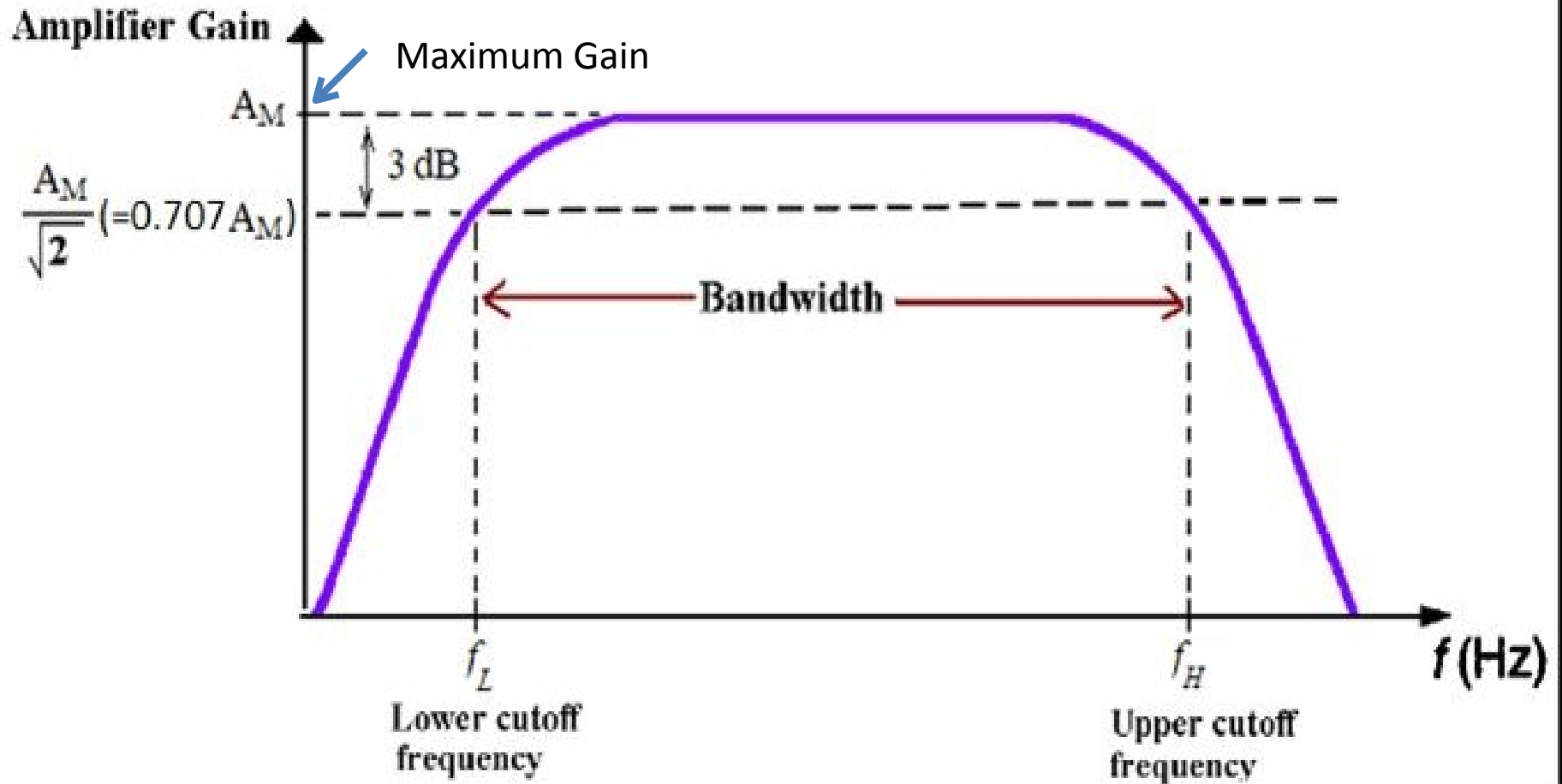
# Frequency Response

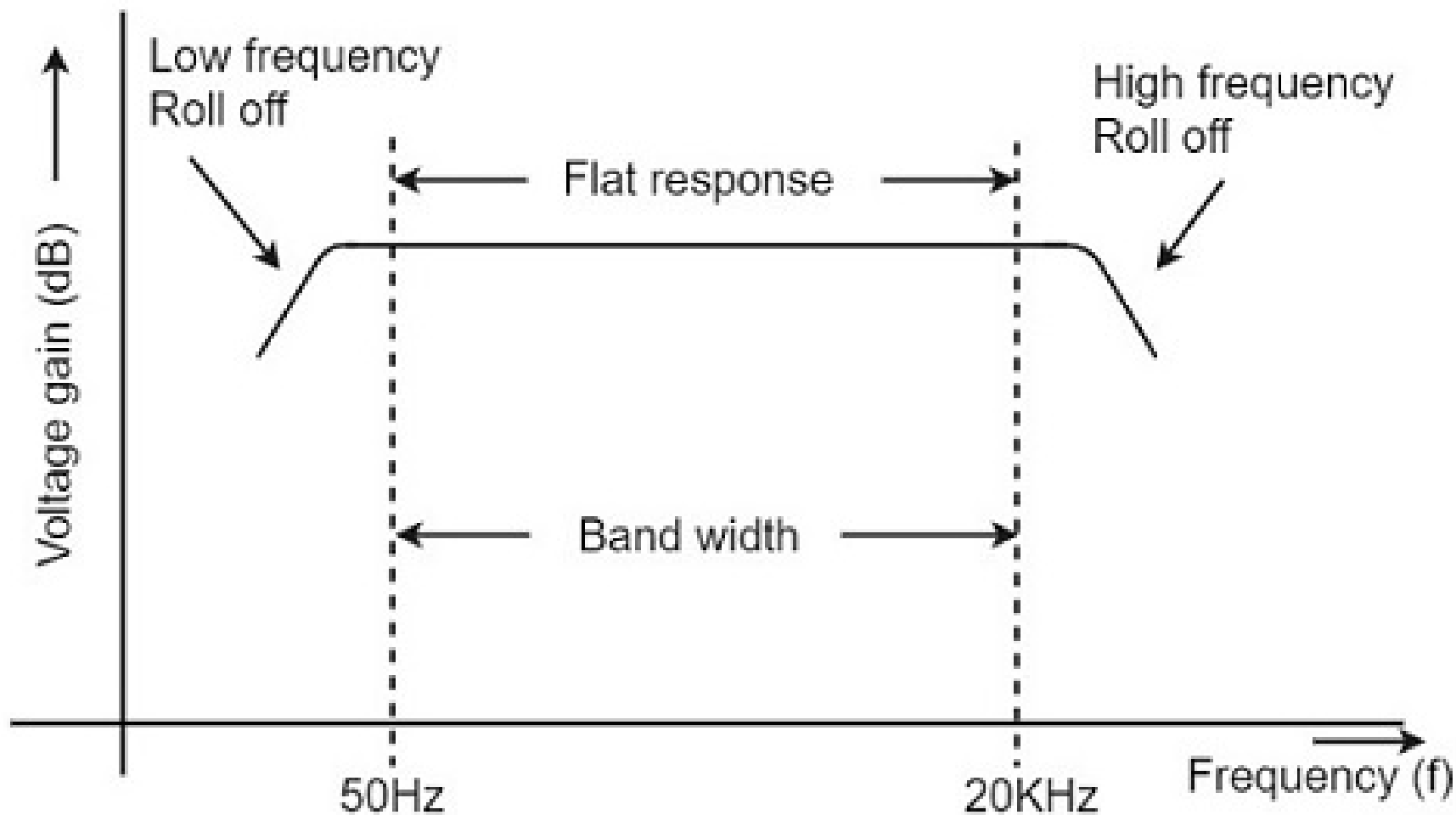




# Frequency Response

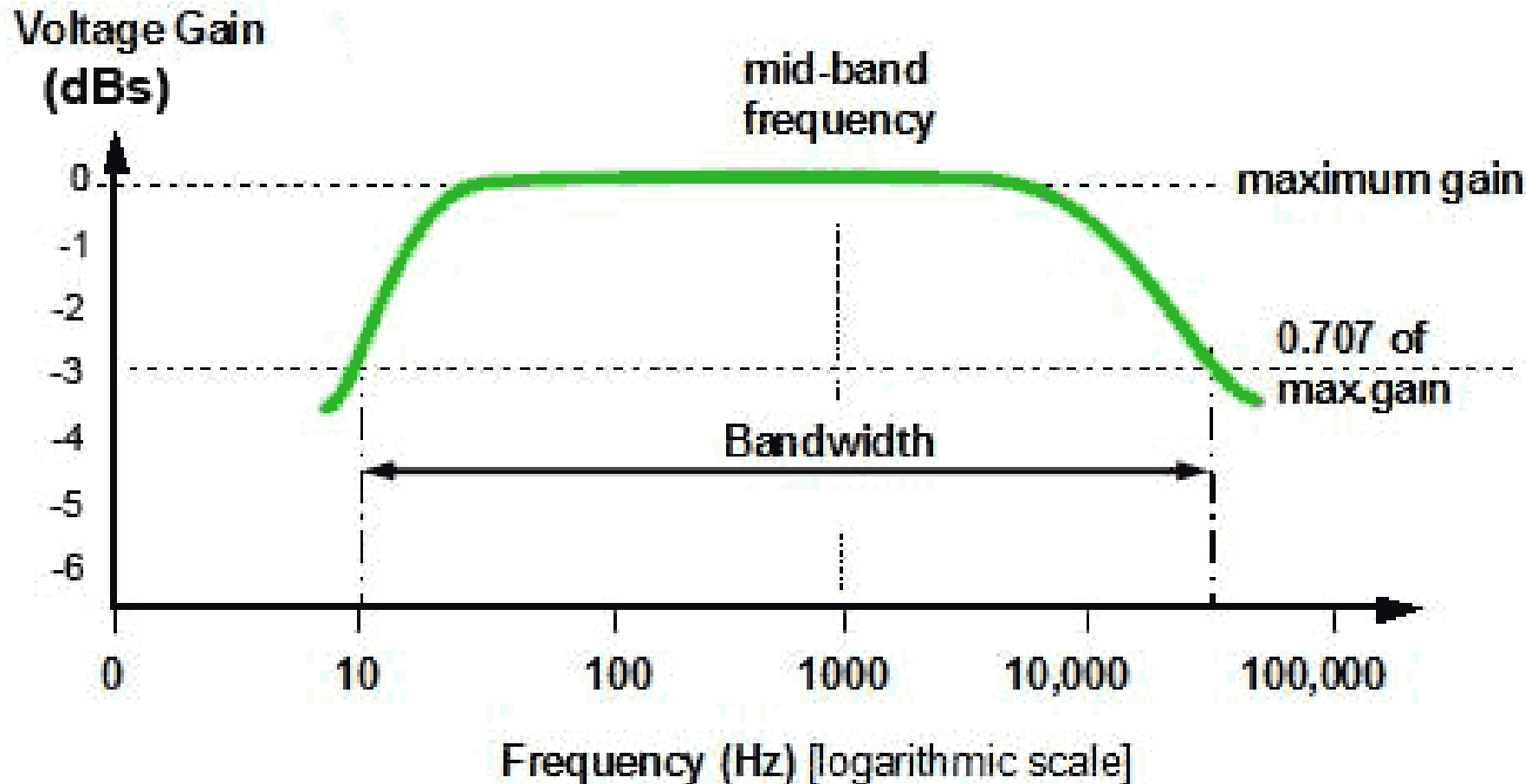
The Frequency Response of an Amplifier





# Frequency Response

X-axis in  $\log f$  and Y-axis in dB



# RC coupled Amplifier

(i) *At low frequencies* ( $< 50$  Hz), the reactance of coupling capacitor  $C_C$  is quite high and hence very small part of signal will pass from one stage to the next stage. Moreover,  $C_E$  cannot shunt the emitter resistance  $R_E$  effectively because of its large reactance at low frequencies. These two factors cause a falling of voltage gain at low frequencies.

# RC coupled Amplifier

(ii) *At high frequencies* ( $> 20$  kHz), the reactance of  $C_C$  is very small and it behaves as a short circuit. This increases the loading effect of next stage and serves to reduce the voltage gain. Moreover, at high frequency, capacitive reactance of base-emitter junction is low which increases the base current. This reduces the current amplification factor  $\beta$ . Due to these two reasons, the voltage gain drops off at high frequency.

# RC coupled Amplifier

(iii) *At mid-frequencies (50 Hz to 20 kHz), the voltage gain of the amplifier is constant. The effect of coupling capacitor in this frequency range is such so as to maintain a uniform voltage gain. Thus, as the frequency increases in this range, reactance of  $C_c$  decreases which tends to increase the gain. However, at the same time, lower reactance means higher loading of first stage and hence lower gain. These two factors almost cancel each other, resulting in a uniform gain at mid-frequency.*

# RC coupled Amplifier

## Advantages

- (i) It has excellent frequency response. The gain is constant over the audio frequency range which is the region of most importance for speech, music etc.
- (ii) It has lower cost since it employs resistors and capacitors which are cheap.
- (iii) The circuit is very compact as the modern resistors and capacitors are small and extremely light.

## Disadvantages

- (i) The *RC* coupled amplifiers have low voltage and power gain. It is because the low resistance presented by the input of each stage to the preceding stage decreases the effective load resistance ( $R_{AC}$ ) and hence the gain.
- (ii) They have the tendency to become noisy with age, particularly in moist climates.
- (iii) Impedance matching is poor. It is because the output impedance of *RC* coupled amplifier is